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Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business

von

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Abstract

This thesis examines how decision-making under uncertainty in football can inform strategic and operational decision-making in business. Both contexts involve rapid judgment under pressure, limited information, and constantly changing conditions. Football offers a visible and structured environment where uncertainty and adaptability are tested in real time. By studying twenty competitive matches between 2014 and 2023, this research builds a framework that connects tactical intelligence in sport with managerial decision-making in business.

The study focuses on identifying turning points such as goals, substitutions, red cards, and tactical changes, treating each as a decision event that shapes outcomes. These events are interpreted qualitatively using structured data extracted from Wyscout reports. Each football scenario is then mapped to an equivalent business situation such as crisis response, leadership under pressure, or recovery after setbacks. Interviews with experienced professionals from relevant industries are used to validate these mappings and ensure practical relevance in business contexts.

The methodology draws from Decision-Making under Deep Uncertainty (DMDU) approaches including Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT). These frameworks emphasize flexibility, adaptability, and learning instead of prediction and control, reflecting the same qualities that define tactical choices in football. An efficiency score is also introduced to assess the quality of decisions in both domains, calculated as the number of effective decisions divided by the total number of decisions.

The research delivers a Decision Intelligence Transfer Framework (DITF) showing how football-based insights, situational awareness, adaptive tactics, and recovery mechanisms can improve decision-making in uncertain business environments. The findings emphasize that effective decisions rely more on context, timing, and communication than on prediction. This work demonstrates how high-pressure conditions in sport can inform business strategy and leadership.

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ALCM	Asset, Liability and Capital Management
DAP	Dynamic Adaptive Planning
DAPP	Dynamic Adaptive Policy Pathways
DITF	Decision Intelligence Transfer Framework
DMDU	Decision Making under Deep Uncertainty
EOA	Engineering Options Analysis
IGDT	Info-Gap Decision Theory
QCA	Qualitative Comparative Case Analysis
RDM	Robust Decision Making
RPD	Recognition-Primed Decision
RQ	Research Question
xG	Expected Goals
XAI	Explainable artificial intelligence

1 Introduction

Decision-making under uncertainty defines both sport and business. Whether a manager adjusts tactics during a football match or a leader responds to a market disruption, both act in environments where information is incomplete and time is limited. Each decision carries risk which depends on situational awareness and often determines success or failure. Football provides a rare and structured view of this process because every match captures real-time decisions, reactions, and consequences within a clearly defined framework of rules and objectives. The same characteristics exist in high-stakes business situations, although they appear less visible and less predictable.

This study builds on that connection. It uses football as a practical lens to understand how individuals and teams make critical choices when outcomes cannot be predicted with certainty. The intent is not to compare sport and business in terms of goals or values, but to learn how tactical intelligence, adaptation, and recovery from mistakes in football can inform business decision-making. By drawing parallels between match dynamics and corporate responses to change or crisis, the study explores what makes certain decisions more resilient and efficient than others.

The research is qualitative in nature. It examines selected football matches as individual cases where uncertainty shaped outcomes and then maps these cases to real-world business equivalents. The approach combines structured event data from professional football analysis platforms with interviews from experienced decision-makers to ensure practical relevance. The result is a framework that captures how effective decision-making relies less on prediction and more on adaptability, coordination, and the ability to respond intelligently to unexpected change.

1.1 Research Context and Motivation

Modern organizations operate in environments where uncertainty has become a structural condition rather than a temporary disruption. Leaders make decisions amid volatile markets, shifting regulations, digital transformations, and unpredictable global events. Traditional rational models of management assume that decision-makers possess reliable data, stable preferences, and the ability to calculate optimal outcomes. In reality, the information is often incomplete, ambiguous, and time sensitive. As John Kay and Mervyn

King argue in Radical Uncertainty: Decision-Making Beyond the Numbers (2020), most strategic problems do not allow for precise probability estimates because the range of possible futures cannot be known in advance.[1] Under such conditions, organizations need decision systems that prioritize adaptability and learning rather than prediction and control.

The field of Decision Making under Deep Uncertainty (DMDU) emerged as a response to these limitations of probabilistic thinking. It provides a set of approaches designed for environments where models, data, and outcomes are contested or unknown. Methods such as Robust Decision Making (RDM), Dynamic Adaptive Planning (DAP), and Dynamic Adaptive Policy Pathways (DAPP) replace the idea of optimization with robustness and flexibility.[2] Instead of relying on a single forecast, these frameworks stress-test multiple strategies against thousands of plausible futures and identify policies that remain viable across a wide range of scenarios. The underlying philosophy is that the quality of a decision should be judged by how well it performs under surprise, not how well it fits a specific prediction.

This shift from prediction to adaptation also draws attention to the human side of decision-making. Even in data-rich settings, choices are filtered through experience, bias, and bounded rationality. Gary Klein's Recognition-Primed Decision (RPD) model, developed through field research with firefighters and military commanders, shows that experts make rapid decisions not by comparing options but by recognising patterns that match prior experience.[3] Similarly, Reale et al. find that in high-risk environments such as aviation or healthcare, individuals rely on a combination of intuitive and analytical reasoning, selecting the first feasible solution under time pressure rather than the theoretically optimal one.[4] These insights confirm that effective decision-making in uncertainty depends as much on experience and feedback as on data.

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[1] See *Kay, J., King, M.*, Radical Uncertainty: Decision-Making Beyond the Numbers, 2020, pp. 48-49.

[2] See *Marchau, V. A. W. J.; Walker, W. E.; Bloemen, P. J. T. M.; Popper, S. W.*, Decision Making under Deep Uncertainty, 2019, p. 17.

[3] See *Klein, G. A.*, Recognition-Primed Decision Model, 1993, p. 139.

[4] See *Reale, C.; Salwei, M. E.; Militello, L. G.; Weinger, M. B.; Burden, A.; Sushereba, C.; Torsher, L. C.; Andreae, M. H.; Gaba, D. M.; McIvor, W. R.; Banerjee, A.; Slagle, J.; Anders, S.,* Decision-Making During High-Risk Events, 2023, pp. 2-3.

Organizational crises show the same dynamics. During the 2008 financial collapse or the early stages of the COVID-19 pandemic, decision-makers were forced to act with incomplete information, under stress, and without consensus about causal models. Those who succeeded were often not the ones with perfect analysis but those who could re-orient quickly when assumptions failed. In this sense, uncertainty management becomes a continuous cycle of observing, interpreting, deciding, and acting, a process captured in John Boyd's OODA loop.[5] Organizations that shorten this loop by learning faster gain a competitive edge.

This thesis situates football as a real-world laboratory for examining decision-making under uncertainty. Every match unfolds in conditions of imperfect information, shifting constraints, and emotional intensity. Coaches and players must constantly process cues, anticipate opponents, and adapt tactics as the game evolves. A red card, an injury, or a change in weather can instantly alter probabilities and priorities. Decisions must balance immediate survival with long-term objectives, just as managers in volatile markets must trade off short-term risk against strategic resilience. Football therefore offers a compressed and observable environment to study how adaptive intelligence emerges in practice.

However, this study does not aim to transfer tactical methods from football into business practice. Instead, it examines whether the strategic reasoning patterns visible in football, particularly under crisis and uncertainty mirror the cognitive and organizational responses observed in business decision-making. The intent is to identify structural equivalence in how humans interpret, adapt, and act when faced with volatility.

The analogy between football and business is not superficial. Both involve complex systems where strategy interacts with chance, and where success depends on reading uncertainty faster than the opponent. Just as a coach must decide whether to substitute early to change momentum, a corporate leader must decide whether to pivot strategy

[5] See *Ryder, M.; Downs, C.*, John Boyd's OODA Loop, 2022, p. 5.

before market conditions worsen. In both settings, the timing, feedback, and adaptability of decisions determine performance outcomes.[6]

By analysing football's tactical decision processes as structured responses to uncertainty, this study explores whether comparable reasoning patterns exist in business contexts. Football condenses the uncertainty of complex systems into a time-bound, high-pressure simulation where every decision has measurable effects. Understanding how uncertainty is perceived, interpreted, and acted upon in this environment can inform the design of business decision systems that are not only data-driven but context-sensitive, resilient, and adaptive.

1.2 Research Objectives and Questions

The purpose of this thesis is to investigate how decision-making under uncertainty can be understood, modelled, and transferred between domains that operate in volatile and information-scarce environments. Traditional rational models often fail when probabilities cannot be assigned, or outcomes predicted with confidence. Frameworks such as Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT) offer tools for developing flexible and adaptive strategies rather than relying on static forecasts.[7] [8] [9]

Football represents a natural experiment in uncertainty. Every match evolves through shifting momentum, unpredictable outcomes, and dynamic feedback loops that mirror business environments facing market shocks or crises.[10]

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6 See *Gabrys, T.; Chruscinski, R.; Garnys, M.; Suva, M.; Švátora, K.*, The Influence of Substitution Decisions Made by National Team Coaches on Final Match Outcomes at UEFA EURO 2024, 2025, p. 2.

7 See *Lempert, R. J.,* Robust Decision Making, 2019, p. 25.

8 See *Haasnoot, M.; Warren, A.; Kwakkel, J. H.*, Dynamic Adaptive Policy Pathways, 2019, p. 72.

9 See *Ben-Haim, Y.*, Info-Gap Decision Theory, 2006, p. 95.

10 See *Horney, N.*, Initiating Action: Applying the OODA Loop to Accelerate Your Decision Making, p. 2.

This thesis formulates five research questions to guide analysis and interpretation:

RQ1. How can football match data be structured and modelled to simulate decision-making under uncertainty?

Theoretical Dimension: Draws on principles from RDM and DAPP to identify how complex systems can be represented through adaptive simulation.[11] [12]

Analytical Dimension: Uses event-level football data to design qualitative simulations that capture uncertainty through variables such as substitutions, tactical changes, and momentum shifts.

RQ2. What types of decisions in football matches reflect real-world high-stakes business decisions?

Theoretical Dimension: Builds on Recognition-Primed Decision (RPD) theory, where intuition and pattern recognition guide rapid judgment under time pressure.[13]

Analytical Dimension: Maps player and coach responses to situational triggers (e.g., conceding a goal) to leadership behaviour in corporate crisis scenarios.

RQ3. How can critical turning points (e.g., goals, tactical changes, game state shifts) be used to analyse decision impact in simulated environments?

Theoretical Dimension: Applies Info-Gap Decision Theory to evaluate robustness and adaptability across alternative.[14]

Analytical Dimension: Uses qualitative coding of match events to measure the effect of decisions on team performance resilience.

RQ4. What methods can be used to build a transferable decision intelligence framework from sports data to business domains?

Theoretical Dimension: Anchored in analogical reasoning, where structural relationships between two domains enable cross-context transfer of insights.[15]

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[11] See *Lempert J.R.*, Robust Decision Making, 2019, p. 25.

[12] See *Haasnoot et al*., Dynamic Adaptive Policy Pathways, 2019, pp. 72 - 89.

[13] See *Klein*, *G.*, Recognition-Primed Decision Model, 1993, p. 139.

[14] See *Ben-Haim*, Info-Gap Decision Theory, 2019, p. 95.

[15] See *Lee, A.*, Analogical Reasoning, 1993, p. 2.

Analytical Dimension: Constructs mappings between tactical intelligence in football and strategic adaptability in business decision models.

RQ5. How can insights from tactical decision modelling in sports enhance decision-support systems in industries like operations, consulting, or crisis management?

Theoretical Dimension: Draws from the Decision Intelligence paradigm, integrating explainable AI and adaptive learning.[16]

Analytical Dimension: Applies findings from football analytics such as event prediction, cognitive modelling, and feedback loops to improve transparency and responsiveness in business decision-support tools.

Collectively, these questions link decision theory, behavioural science, and sports analytics to develop a framework for adaptive intelligence under deep uncertainty.

1.3 Theoretical Framing

This study builds on a dual theoretical framework that combines decision science with analogical reasoning. The first pillar focuses on decision-making under deep uncertainty (DMDU) and its key methodological families such as Robust Decision Making (RDM), Dynamic Adaptive Planning (DAP), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT). The second pillar uses analogy logic to explore how reasoning patterns visible in football decision-making parallel those found in business contexts. Together, these perspectives support a framework that interprets rather than transfers tactical intelligence, enabling the identification of shared cognitive and structural features across domains.

The concept of deep uncertainty is central to modern decision science. Marchau, Walker, Bloemen, and Popper define it as a condition where stakeholders cannot agree on the relevant models, probabilities, or outcomes of decisions.[17] In such contexts, predictive or optimization-based strategies become unreliable. RDM and its associated methods emerged as a response to this limitation, emphasizing robustness and adaptability over precision. Lempert’s formulation of RDM shifts attention from finding an “optimal”

[16] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, p. 375.

[17] See *Marchau et al*., Decision Making under Deep Uncertainty: From Theory to Practice, 2019 p. 15.

decision to identifying strategies that perform consistently across a wide range of plausible futures.[18]

Dynamic Adaptive Planning (DAP) and Dynamic Adaptive Policy Pathways (DAPP) expand on this logic by viewing decisions as evolving sequences rather than static choices. DAP introduces the use of signposts and triggers that indicate when a plan should be adjusted in response to changing conditions.[19] DAPP builds on this by designing multi-pathway frameworks that visualize how different strategies can branch and rejoin as contexts evolve.[20] Info-Gap Decision Theory complements these by introducing a formal measure of robustness to uncertainty. Rather than estimating probabilities, it evaluates how much deviation from expectation a decision can withstand before failing.[21]

While these frameworks were developed primarily for policy, climate adaptation, and infrastructure planning, their logic extends naturally to domains such as sport and business, which also feature feedback loops, time constraints, and complex interdependencies. Popper's reflections on DMDU emphasize that decisions in such systems must integrate analytical foresight with iterative learning and stakeholder engagement.[22]

The second pillar, analogy logic, provides the interpretive bridge between these decision frameworks and football. Bartha defines analogical reasoning as a process of mapping relations between two systems to transfer insight or structure from a familiar source to a less familiar target.[23] In this context, the football field becomes an experimental setting for understanding uncertainty management, where fast, high-stakes decisions mirror the psychological and structural pressures of business environments.

However, this study does not attempt to transplant tactical methods from football into business decision systems. It applies analogy as a comparative reasoning tool, a structured

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[18] See *Lempert, R. J., Robust* Decision Making, 2019, p. 25.

[19] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 55-56.

[20] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 73-74.

[21] See *Ben-Haim*, Info-Gap Decision Theory, 2019, pp. 96-97.

[22] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, p. 387.

[23] See *Bartha, P.,* Analogical Reasoning, 2019, p. 12.

method for analysing whether the cognitive and adaptive mechanisms used by coaches and players resemble those used by managers and executives. Sinnicks warns that sport-business analogies risk distortion if they ignore contextual and ethical boundaries; sporting competition is voluntary and internally regulated, whereas business actions can affect external stakeholders.[24] By maintaining contextual integrity, analogy in this thesis functions as a disciplined comparison rather than a direct import.

By combining decision science and analogy logic, this theoretical framing allows a dual perspective: DMDU offers the analytical language to describe decision environments, and football provides a real-time setting for observing adaptive cognition under stress. The framework ultimately explores how humans interpret signals, adapt to evolving feedback, and sustain performance across uncertainty, not to replace business tools with sporting strategies, but to uncover the shared logic that underlies both.

1.4 Structure of the Thesis

This thesis is organized into seven main chapters, moving from theoretical foundations to applied analysis and synthesis. The progression follows a logical structure that mirrors the research process itself, beginning with conceptual framing, moving through empirical exploration, and concluding with an integrative framework for decision intelligence.

Chapter 1 introduces the topic, outlines the research motivation, and defines the theoretical and analytical structure that underpins the study. It situates the problem of decision-making under uncertainty within organizational and sporting contexts and identifies football as a controlled yet dynamic setting for observing adaptive intelligence.

Chapter 2 presents the literature review. It draws from established decision science frameworks, including Robust Decision Making, Dynamic Adaptive Policy Pathways, and Info-Gap Decision Theory, while also incorporating research in sports analytics, cognitive modelling, and analogical reasoning. This chapter builds the intellectual bridge between theory and the football-based application.

Chapter 3 details the research design and methodology, explaining how the 20 selected matches and five semi-structured interviews were analysed qualitatively. It also describes

[24] See *Sinnicks, M.*, Business-Sport Analogy Critique, 2022, p. 50.

the coding matrix and analytical process developed for pattern recognition and theme extraction.

Chapters 4 and 5 present the core analysis. The matches are grouped into four thematic clusters namely Crisis and Recovery, Tactical Adaptation, Leadership Under Pressure, and Controversy and Unpredictability, each linked to specific decision-making concepts derived from DMDU theory.

Chapter 6 discusses cross-case patterns and proposes the Decision Intelligence Transfer Framework (DITF), which integrates the findings from football and business contexts.

Chapter 7 concludes the thesis by summarizing key insights, theoretical implications, and directions for future research.

Together, these chapters form a cohesive argument that demonstrates how decision-making frameworks developed for uncertainty in policy and engineering can be extended into the social and cognitive dimensions of sport and business.

2 Literature Review

This chapter reviews the core bodies of literature that inform this thesis. It begins by examining decision-making under uncertainty, with a particular focus on frameworks developed for conditions of deep uncertainty. It then reviews relevant research in sports analytics and cognitive modelling, followed by literature that uses sport as an analogue for business and strategic decision-making. The chapter concludes by identifying the conceptual and theoretical gap that this thesis addresses.

2.1 Decision-Making Under Uncertainty

Decision-making under uncertainty is one of the most studied yet least resolved challenges in both theory and practice. Traditional decision frameworks have long relied on probabilistic models and optimization techniques, assuming that future outcomes can be predicted within known ranges of probability. However, as Marchau, Walker, Bloemen, and Popper (2019) argue, real-world contexts often operate under conditions of deep uncertainty, where stakeholders cannot agree on models, probabilities, or even the possible set of future outcomes.[25] In such environments, prediction-based planning not only fails to capture complexity but also leads to rigid strategies that collapse when exposed to unforeseen events.

The field of Decision-Making under Deep Uncertainty (DMDU) emerged as a response to these limitations. Its methods like Robust Decision Making (RDM), Dynamic Adaptive Planning (DAP), Dynamic Adaptive Policy Pathways (DAPP), Info-Gap Decision Theory (IGDT), and Engineering Options Analysis (EOA) represent a shift away from prediction and optimization toward robustness, flexibility, and learning. These frameworks do not assume a single "best" course of action but rather seek strategies that remain viable across a range of plausible futures.

Among these, Robust Decision Making (RDM) introduced by Lempert stands as a foundational approach. RDM operates through simulation-based scenario discovery, where a large ensemble of potential futures is used to test how different strategies perform under varying assumptions.[26] The emphasis is not on optimization but on minimizing

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[25] See *Marchau et al.*, Decision Making under Deep Uncertainty: From Theory to Practice, 2019, pp. 15-16.

[26] See *Lempert, R. J., Robust* Decision Making, 2019, p. 25.

regret the vulnerability of a strategy to poor performance in adverse conditions. Lempert's approach allows decision-makers to visualize which policies remain acceptable across uncertainty ranges and which ones fail under stress. This probabilistic detachment marks a conceptual break from traditional decision theory, where confidence in forecasts was mistaken for control.

Dynamic Adaptive Planning (DAP) builds upon this logic by operationalizing adaptability. Walker, Marchau, and Kwakkel describe DAP as a framework that establishes signposts and triggers observable indicators that signal when a plan must be adjusted.[27] This makes decision-making a dynamic process rather than a one-time act. DAP acknowledges that information changes over time and that the capacity to revise a decision is often more valuable than the initial decision itself.

Dynamic Adaptive Policy Pathways (DAPP) extends this thinking even further by visualizing how decisions evolve along multiple pathways. Haasnoot, Warren, and Kwakkel define DAPP as a structured process that maps adaptation tipping points, the moments when a given policy no longer meets its objectives.[28] The emphasis on visualization and stakeholder dialogue makes DAPP an especially useful tool for communicating uncertainty to non-technical audiences. The approach encourages a mindset of continuous learning and preparedness rather than static risk management.

In parallel, Info-Gap Decision Theory (IGDT), developed by Ben-Haim (2019), introduces a mathematically rigorous method for evaluating robustness under severe uncertainty. Instead of focusing on probability distributions, Info-Gap assesses how far a decision can deviate from expectations before it fails to meet acceptable performance criteria.[29] This approach redefines success as "satisfactory performance despite unknowns," reflecting a philosophical stance that treats uncertainty not as a flaw in knowledge but as a natural feature of complex systems.

Engineering Options Analysis (EOA), introduced by de Neufville and Smet, adapts the concept of real options from finance to engineering and infrastructure contexts. EOA quantifies the value of flexibility the ability to defer, expand, or switch investments in

[27] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 53-54

[28] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 72-73.

[29] See *Ben-Haim*, Info-Gap Decision Theory, 2019, p. 96.

response to emerging information.[30] By embedding optionality into project design, organizations can preserve adaptability and avoid path dependency. This logic resonates with DAP and DAPP but offers a financial valuation mechanism that strengthens institutional buy-in.

Each of these DMDU methods shares a common philosophy: that decision-making in complex, uncertain environments must prioritize resilience, learning, and modularity. Popper emphasizes that this orientation toward "deliberation with analysis" combines quantitative scenario exploration with transparent, participatory decision-making.[31] The focus shifts from trying to predict one future to preparing for several possible futures. This is a change in how we think about knowledge and how we make sense of situations, which has important effects on policy, business, and sport.

Critiques of probabilistic and predictive thinking reinforce this reorientation. Kay and King describe uncertainty as radical rather than measurable, arguing that probabilities cannot represent unknown unknowns.[32] They highlight that real-world decision-makers do not calculate probabilities but build narrative rationality, stories that help them navigate ambiguity. Similarly, Stirling and Taleb's perspectives challenge the illusion of statistical precision. Taleb's Black Swan thesis shows that outlier events dominate outcomes and invalidate probabilistic expectations, while Stirling's work on risk governance underscores the necessity of plural and adaptive approaches.[33], [34] These critiques align directly with DMDU, which does not attempt to assign false certainty to inherently unknowable systems.

Empirical studies from other domains confirm that under high stress and time pressure, decision-making departs from rational optimization. Reale et al. found that experts in high-risk environments such as aviation, healthcare, and emergency response rely on a shifting mix of recognition-primed intuition, heuristics, and adaptive learning rather than

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[30] See *Neufville, R.; Smet, K.*, Engineering Options Analysis, 2019, pp. 118-119.

[31] *See Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, p. 389.

[32] See Kay, J.; King, M., Radical Uncertainty, 2020, p. 18.

[33] See *Taleb, N.*, The Black Swan, 2007, p. 1.

[34] See *Stirling, A.*, Risk and Ambiguity, 2003, cited in *Renn, O.; Klinke, A.*, Risk Governance and Resilience, 2014, p. 6.

formal analysis.[35] The finding reinforces that in complex systems, the capacity to adapt in real time matters more than adherence to pre-set rules.

From a conceptual standpoint, this synthesis leads to what can be termed adaptive intelligence, a mindset that integrates the analytical structure of DMDU with the behavioural flexibility observed in real-world decision environments. Adaptive intelligence acknowledges that no framework can eliminate uncertainty; instead, it focuses on sensing, interpreting, and responding effectively as situations evolve. It combines the robustness of DMDU with the intuition and learning described in cognitive decision theories like Klein's Recognition-Primed Decision Model.[36]

The development of adaptive intelligence marks a convergence between technical and cognitive schools of thought. On one side, RDM, DAP, and DAPP provide the structural tools for designing robust and flexible systems. On the other, behavioural research in high-stakes decision-making shows how individuals operate within those systems. This intersection forms the foundation for the present study's analytical model: by observing adaptive intelligence in football, a domain defined by constant uncertainty, we can extract insights applicable to organizational contexts where similar decision pressures exist.

In summary, the literature on decision-making under uncertainty points toward a paradigm that values flexibility, learning, and robustness over prediction and control. From the conceptual advances of DMDU to the behavioural realism of high-stakes cognition, the emphasis is consistent: uncertainty cannot be eliminated, but it can be managed through adaptive intelligence. This synthesis provides the theoretical foundation for analysing football as a live simulation of decision-making under uncertainty and for transferring those insights into business and organizational practice.

2.2 Sports Analytics and Cognitive Modelling

Football offers a unique lens through which complex decision-making can be examined in real time. Each match unfolds within an environment defined by uncertainty, incomplete information, and limited reaction time. Players, coaches, and analysts are constantly required to assess evolving situations and make rapid tactical choices. The

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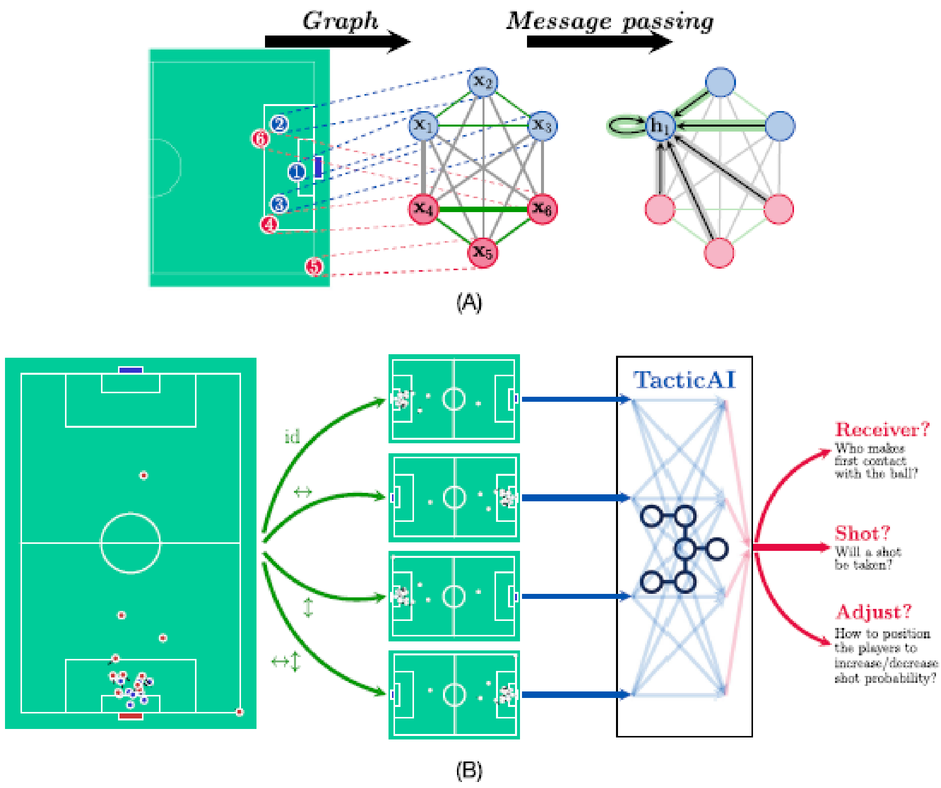
[35] See *Reale et al.*, Decision-Making During High-Risk Events, 2023, p. 3.

[36] See *Klein, G.*, Recognition-Primed Decision Model, 1993, pp. 141-143.

interaction between structured data, human cognition, and adaptive reasoning has turned football into a living case study for understanding how decision-making functions under pressure and ambiguity.

Modern analytics treat football not as a sequence of isolated events but as a continuous process of prediction and adaptation. TacticAI (2024) presents this view by showing how artificial intelligence models can analyse millions of match events to simulate alternative outcomes and tactical responses.[37]

Figure 1: Graph-based representation of corner kick tactics using TacticAI.



Source: *Wang et al.*, TacticAI, 2024, p. 2.

Figure 1: This figure illustrates how TacticAI converts corner-kick situations into graph-based representations to simulate alternative outcomes and tactical responses. By modelling players as interconnected nodes and updating their interactions through

[37] See *Wang, Z. et al.*, TacticAI, 2024, p. 2.

message passing, the system captures football not as isolated events but as a continuous process of prediction and adaptation aligning with the analytical approach used in this thesis.

Such modelling reflects the same reasoning found in Robust Decision Making (RDM), which tests strategies across many potential futures instead of relying on a single forecast.[38] On the pitch, this principle is visible in the constant adjustments that managers and players make based on unfolding information. The study The Influence of Substitution Decisions at UEFA EURO 2024 (2024) showed that the timing and type of substitutions often determine match outcomes, especially in knockout games where time and uncertainty intensify.[39] These choices rely on both measurable data, such as player fatigue and position metrics, and non-measurable elements like momentum or perceived pressure. These decisions often involve balancing measurable indicators, such as player fatigue or positional metrics, with less tangible factors like momentum or emotional stability. In such situations, decision-making is guided by what is workable in the moment rather than by identifying a single optimal choice. The focus becomes identifying an effective response that maintains stability and control under pressure, even when full information is not available.

Cognitive models provide further insight into how expert decision-making operates in football. Klein’s Recognition-Primed Decision (RPD) model explains that experts recognize familiar patterns in evolving situations and select the most workable response without analysing every alternative.[40] This model is particularly relevant in football, where players must act within milliseconds. For instance, midfielders scan the field, recognize opponent structures, and anticipate passing lanes before they even receive the ball. Their responses are not impulsive but informed by thousands of repetitions that create what can be described as intuitive expertise.

This form of rapid cognition parallels findings from Reale et al. (2023), who examined decision-making in high-risk environments such as aviation, healthcare, and crisis

[38] See *Lempert, R. J., Robust* Decision Making, 2019, pp. 26-28.

[39] See *Gabrys, T.; Chruscinski, R.; Garnys, M.; Suva, M.; Švátora, K.*, Substitution Decisions at UEFA EURO 2024, 2025, pp. 1-2.

[40] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 143-144.

response. The authors observed that professionals under pressure combine analytical reasoning with experience-based intuition to balance accuracy and speed.[41] Similar dynamics can be found in football where the tempo of the game leaves no time for step-by-step reasoning, yet the outcomes depend heavily on mental adaptability.

As data collection in sport has become more sophisticated, analytical modelling has shifted from basic statistics to predictive systems. Event Detection in Football (2023) introduced algorithms capable of identifying tactical phases, formation changes, and pressing patterns by analysing video and sensor data.[42] This type of event modelling echoes the idea of signposts and adaptation triggers found in Dynamic Adaptive Planning, where systems respond to specific indicators of change.[43] In football, these triggers might include a sudden drop in pressing efficiency, the loss of numerical superiority in midfield, or the impact of a red card.

Bayesian inference has also become an essential tool in sports analytics. It allows analysts to update probabilities as new data arrives, mirroring the way decision-makers revise beliefs in uncertain contexts. The paper A Predictive Analytics Model for Forecasting Outcomes in the NFL demonstrated how Bayesian models could recalibrate win probabilities after each possession.[44] In football, similar models can continuously evaluate the likelihood of scoring based on ball location, pressure, and team structure. This incremental updating process represents a form of learning that parallels Dynamic Adaptive Planning (Walker et al., 2019), where policy adjustments are made as new information becomes available.[45]

Reinforcement learning extends this concept further. By simulating repeated interactions between agents and their environment, reinforcement learning models enable virtual teams to improve performance through trial and error. Mapping Football Tactical Behaviour and Collective Dynamics with Artificial Intelligence (2023) illustrates how reinforcement learning can be used to reproduce coordinated pressing strategies and

[41] See *Reale et al.*, Decision-Making During High-Risk Events, 2023, pp. 7-8.

[42] See *Bischofberger, J.; Baca, A.; Schikuta, E.*, Event Detection in Football, 2024, pp. 2-3.

[43] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 54-56.

[44] See *Gifford, M.; Bayrak, T.*, Predictive Analytics in the NFL, 2023, p. 2.

[45] See *Walker et al.*, Dynamic Adaptive Planning, 2019, p. 64.

positional structures.[46] This is conceptually similar to the adaptive learning cycles proposed in DMDU frameworks, where feedback and monitoring drive continuous improvement.

Cognitive load theory explains how experts cope with this volume of real-time information. The paper Street-Smart Decision-Making in Sport (2023) found that experienced players and coaches simplify complex decision problems by using pre-learned scripts and cues that reduce the need for conscious computation.[47] This approach allows them to act quickly without losing accuracy. Instead of processing every variable, they focus on the few indicators that matter most for the decision at hand. This reflects the core principle of adaptive intelligence: the ability to distinguish between critical and irrelevant information in uncertain conditions.

A growing branch of research focuses on explainable artificial intelligence (XAI), which aims to make machine reasoning understandable to human users. Explainable AI in Football (2024) demonstrates how transparency in model design can enhance trust and collaboration between analysts and coaching staff.[48] Rather than producing opaque predictions, explainable models show the reasoning process behind tactical recommendations. This is consistent with Popper's notion of "deliberation with analysis," where decisions are supported by analytical evidence that remains interpretable to stakeholders.[49]

The intersection of human cognition and computational modelling forms the foundation for adaptive intelligence in sport. Decision trees, Bayesian inference, and reinforcement learning provide the analytical tools, while cognitive theories such as bounded rationality and recognition-primed intuition explain how humans interpret and apply these tools in practice. Football demonstrates how these elements coexist in a feedback loop where experience shapes perception, perception guides decision, and outcomes refine future action.

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[46] See *Teixeira, J.E. et al.*, Mapping Football Tactical Behaviour, 2025, pp. 2-3.

[47] See *Hecksteden et al.*, Street-Smart Decision-Making in Sport, 2023, pp. 2-3.

[48] See *Vaykole, N.D.*, Explainable AI in Football, 2024, p. 1.

[49] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 388-389.

In essence, sports analytics and cognitive modelling reveal that the most effective decision systems are not purely algorithmic or purely intuitive but an integration of both. The ability to interpret data, recognize patterns, and adapt to evolving circumstances defines not only success on the pitch but also effective decision-making in complex organizational settings. Football, therefore, becomes more than a metaphor for business. It functions as a controlled environment where adaptive intelligence can be observed, measured, and eventually transferred to broader decision-making frameworks.

2.3 Football-to-Business Analogy

The relationship between sport and business has long been discussed as a way to understand leadership, performance, and decision-making under pressure. Football, in particular, offers a structured yet volatile environment where coordination, trust, and adaptability determine outcomes. Managers in both domains face similar challenges: ambiguous information, high stakes, and the need to act without full certainty. The appeal of using football as an analogy for business lies in its clarity. Decisions and their consequences are visible almost immediately, making it a condensed model of organizational behaviour under uncertainty.

At its best, the football-to-business analogy provides a framework for examining how strategy, teamwork, and resilience evolve in dynamic contexts. The literature on analogical reasoning emphasizes that effective analogies depend on the structural similarities between the source and the target domain rather than surface resemblance.[50] In this sense, football functions as a valid analogical source because it exhibits the same systemic features found in business: competition for limited resources, adaptive tactics, and continuous feedback loops.

Within organizational theory, leadership in football is often interpreted through distributed and situational models. Team captains and coaches share responsibilities much like executives and department heads in firms. Leadership is not a single decision but a process of balancing autonomy and coordination. The Street-Smart Decision-Making in Sport study (2023) showed that high-performing teams rely on collective

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[50] See *Stanford Encyclopedia of Philosophy*, Analogy and Analogical Reasoning, Fall 2019 Edition, pp. 3-4.

intelligence, where players make local decisions that align with a shared overall strategy.[51] This mirrors decentralized management systems that empower employees to adapt in real time while maintaining organizational coherence.

Resilience is another transferable concept. In football, resilience manifests through a team's ability to recover from setbacks such as conceding an early goal or losing a key player. In business, it reflects how organizations respond to shocks like market collapse or operational disruption. The DMDU framework, especially the work on Dynamic Adaptive Policy Pathways, formalizes resilience as the capacity to adapt before thresholds are breached.[52] Football teams that modify tactics mid-game when indicators signal vulnerability are demonstrating this same anticipatory resilience.

Teamwork provides a third area of convergence. Both football squads and business units depend on communication, role clarity, and trust. The study Mapping Football Tactical Behaviour and Collective Dynamics with Artificial Intelligence used AI to map passing networks and movement synchronization, showing how smooth teamwork correlates with success.[53] The same principle applies to business networks where inter-departmental coordination and information flow determine overall performance. These parallels are not merely metaphorical; they reveal shared structural dependencies between cooperation, communication, and adaptability.

However, the analogy between football and business is not without criticism. Sinnicks cautions that the frequent use of sport as a metaphor risks romanticizing competition and oversimplifying moral and contextual complexity.[54] He argues that businesses rarely operate under conditions of clear rules or fair play, while sport depends on codified structures that make outcomes legible. Similarly, Gammelsaeter questions whether the hierarchical and performance-driven logic of elite sport can genuinely inform ethical or sustainable leadership in organizations.[55] For these authors, the analogy can distort rather than clarify if used without theoretical precision.

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[51] See *Hecksteden et al.*, Street-Smart Decision-Making in Sport, 2023, pp. 4-6.

[52] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 73-76.

[53]See *Teixeira, J.E. et al.*, Mapping Football Tactical Behaviour, 2025, pp. 2-4.

[54] See *Sinnicks, M.*, Business-Sport Analogy Critique, 2022, pp. 52-53.

[55] See *Gammelsaeter, H.*, Sport is not Industry, 2020, pp. 2-4.

The value of a football analogy therefore lies in disciplined application. The Analogy and Analogical Reasoning framework explains that meaningful transfer requires mapping relational structures, not replicating surface features.[56] When businesses borrow from football, the focus should be on processes learning under pressure, distributed communication, and adaptive strategy rather than imitating sport's competitive rituals or celebrity leadership culture.

From a cognitive standpoint, both football and business environments depend on recognition, anticipation, and situational awareness. The OODA loop, developed by John Boyd, provides a shared cognitive framework: observe, orient, decide, and act.[57] Footballers apply this cycle repeatedly during play, updating mental models with every new stimulus. Executives facing volatile markets follow a similar rhythm when they scan for information, interpret signals, and commit to strategic actions. Both systems reward rapid yet reflective adaptation rather than mechanical execution.

Another parallel arises in the concept of "deliberation with analysis" from Popper, which argues that robust decisions emerge from iterative testing and transparent evaluation.[58] Football analytics operates the same way: tactical systems are validated across multiple match scenarios before being institutionalized. In business, such iterative reasoning underpins evidence-based management and agile decision processes.

Yet the analogy has limits that must be acknowledged. Football's clarity of rules, measurable outcomes, and defined timeframes contrasts with the ambiguity of organizational life. Firms rarely experience full information feedback, nor do they have referees to enforce fairness. Moreover, success in football is zero-sum; one team's victory requires another's defeat. In business ecosystems, collaboration, regulation, and shared infrastructure complicate this binary outcome. Over-reliance on sporting metaphors can therefore obscure the cooperative and ethical dimensions of management that extend beyond performance alone.[59]

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[56] See *Bartha, P.*, Analogical Reasoning, 2019, p. 22.

[57] See *Ryder, M.; Downs, C.*, Rethinking Reflective Practice: John Boyd's OODA Loop as an Alternative to Kolb, 2022, pp. 4-5.

[58] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 388-389.

[59] See *Gammelsaeter, H.*, Sport is not Industry, 2020, pp. 6-7.

Despite these caveats, football remains a valuable interpretive model when used thoughtfully. It exposes how leadership and teamwork operate under uncertainty, how strategies adapt in response to feedback, and how individuals manage cognitive load in time-pressured environments. The football field becomes a living simulation of organizational behaviour where every decision can be observed, analysed, and connected to broader theories of adaptive intelligence.

When combined with the methodological insights of Decision Making under Deep Uncertainty, the analogy gains further depth. Football demonstrates the principles of robustness and adaptability that DMDU advocates: preparing for multiple futures, identifying early warning signs, and maintaining flexibility to adjust course.[60] By framing football decisions as experiments in adaptive strategy, organizations can extract lessons that are both practical and conceptually rigorous.

The potential of this analogy lies not in copying tactics but in transferring mindsets. Football teams succeed when they anticipate uncertainty, respond with cohesion, and learn continuously. Businesses facing volatile markets can adopt similar patterns by embedding feedback loops, empowering decentralized decision-making, and cultivating shared situational awareness. These structural parallels form the foundation of the Decision Intelligence Transfer Framework developed later in this thesis.

In conclusion, the football-to-business analogy offers a powerful but nuanced model for understanding decision-making under uncertainty. Its strength comes from structural similarity, not superficial imitation. Used critically, it illuminates how leadership, resilience, and teamwork evolve in dynamic systems and how adaptive intelligence can move from the sports arena into the organizational world. When misapplied, it risks distortion or moral oversimplification. The task, therefore, is to treat the analogy as an analytical bridge grounded in evidence, aware of its boundaries, and capable of generating new insights about how humans and systems decide when the future cannot be predicted.

2.4 Conceptual Gap

Across the reviewed literature, one recurring limitation stands out: decision-making frameworks and real-time human cognition are often studied separately. The field of

[60] See *Marchau et al.*, Decision Making under Deep Uncertainty, pp. 12-16.

Decision Making under Deep Uncertainty (DMDU) has developed strong analytical tools such as Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT) to structure choices in uncertain environments. These models formalize uncertainty and provide mechanisms for resilience, but they remain primarily policy-oriented and rarely engage with the rapid, instinctive nature of human decisions observed in sport. Conversely, football research has produced rich insights into perception, intuition, and team coordination, yet it lacks an overarching framework that connects cognitive adaptation to structured models of uncertainty. This separation has left a conceptual gap between formal decision science and the dynamic intelligence seen in real-world performance systems.

The first part of this gap arises from methodological abstraction. RDM and related frameworks focus on computational exploration of scenarios to identify strategies that remain robust across multiple futures.[61] While this approach ensures analytical discipline, it does not fully account for the speed and subjectivity of human decision processes. In football or crisis situations, individuals cannot run thousands of simulations before acting. They rely on recognition, pattern recall, and intuition within tight temporal limits. Dynamic Adaptive Planning (DAP) and DAPP address some of this by embedding triggers and signposts into policy design.[62], [63] Yet these mechanisms are still procedural rather than cognitive. They describe what an organization should do, not how a human mind adapts when faced with complexity.

The second part of the gap lies in the cognitive sciences applied to sport. Studies such as Street-Smart Decision-Making in Sport, show that expert performers rely on pattern recognition, simplified heuristics, and anticipatory attention to operate under pressure.[64] Similarly, Klein's Recognition-Primed Decision (RPD) Model demonstrates that individuals can act effectively without comparing multiple options because they recognize familiar patterns in unfolding contexts.[65] These perspectives highlight

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[61] See *Lempert, R. J., Robust* Decision Making, 2019, p. 33.

[62] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, p. 72.

[63] See *Walker et al.*, Dynamic Adaptive Planning, 2019, p. 54.

[64] See *Hecksteden, A. et al.*, Street-Smart Decision-Making in Sport, 2023, p. 12.

[65] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 142-143.

adaptability but rarely connect to macro-level decision frameworks like DMDU or RDM. The result is a divide between micro-cognitive flexibility and macro-strategic robustness.

Several attempts have been made to bridge this divide. Engineering Options Analysis (EOA) quantifies flexibility by assigning value to design options that can expand or switch in the future.[66] While EOA makes flexibility measurable, it still treats decisions as discrete events rather than fluid cognitive sequences. Conversely, models like the OODA loop (Observe, Orient, Decide, Act) developed by Boyd (1987) emphasize real-time adaptation.[67] Yet they remain qualitative and are not formally linked to analytical planning frameworks.

The sports analytics literature offers glimpses of integration. Artificial-intelligence-based models such as TacticAI (2024) and Mapping Football Tactical Behaviour and Collective Dynamics (2023) attempt to merge quantitative analysis with cognitive reasoning by modelling how teams adapt to emerging patterns.[68], [69] However, these models still focus on data interpretation rather than decision reasoning. They predict what is likely to happen next but do not explain why a particular action was cognitively selected. Explainable AI approaches address this issue by making algorithmic reasoning transparent, aligning machine logic with human interpretability.[70] Still, they lack an explicit connection to decision frameworks such as DAPP or RDM, which could ground their findings within a broader structure of adaptive governance.

The third part of the gap is about how we understand knowledge itself. Traditional probability-based thinking assumes that uncertainty can be reduced just by gathering more data or improving models. Critics such as Kay and King, and Stirling argue that this assumption creates an illusion of control, leading to misplaced confidence in

[66] See *Neufville, R.; Smet, K.*, Engineering Options Analysis: Applications, 2019, pp. 231.

[67] See Ryder, M.; Downs, C., Rethinking Reflective Practice: John Boyd's OODA Loop as an Alternative to Kolb, 2022, pp. 7-8.

[68] See *Wang, Z. et al.*, TacticAI, 2024, pp. 2-3.

[69] See *Teixeira, J.E. et al.*, Mapping Football Tactical Behaviour, 2025, pp. 3-5.

[70] See *Vaykole, N.D.*, Explainable AI in Football, 2024, p. 1.

forecasts.[71],[72] Football and other dynamic environments demonstrate the opposite: not all uncertainty can or should be eliminated. Players and managers often make decisions by embracing ambiguity, using feedback to guide continuous adjustment. This perspective aligns with Popper's (2019) interpretation of DMDU as a governance paradigm centred on learning rather than prediction.[73]

What remains missing, therefore, is a coherent bridge between structured and intuitive approaches. On one side, DMDU offers a vocabulary for designing robust and adaptive strategies across long-term horizons. On the other, cognitive science and football analytics reveal how individuals and teams enact adaptability in real time. Both speak to uncertainty, but in different languages. The current thesis positions itself in this unoccupied space by linking formal frameworks of uncertainty management with the lived reality of rapid, feedback-driven decision-making.

Conceptually, this bridge will take the form of a Decision Intelligence Transfer Framework. The framework will align the structural rigor of DMDU methods with the cognitive dynamism of sport. It will treat football matches as natural laboratories for observing adaptive intelligence: systems where decision agents continuously balance data, intuition, and collaboration. By mapping football events to cognitive responses and business analogues, the study will show how real-time adaptation can be translated into structured decision processes without losing flexibility.

The proposed contribution is twofold. Theoretically, it extends the DMDU discourse beyond institutional and policy contexts to include human cognition and team dynamics. Practically, it demonstrates how the same logic of robustness, feedback, and learning that guides adaptive policymaking can inform management decisions in volatile business environments. Football, with its constant interplay of structure and improvisation, provides the bridge that has been missing between analytical frameworks and the human mind under pressure.

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[71] See *Kay. J; Kings. M*, Radical Uncertainty, 2020, p. 329.

[72] See *Stirling, A.*, Risk and Ambiguity, 2003, cited in Renn, O.; Klinke, A., Risk Governance and Resilience, 2014, p. 6.

[73] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 3-5.

In summary, existing research has either emphasised formal modelling at the expense of cognition or focused on intuition without systemic structure. This thesis responds by integrating both. It will use the decision architectures of DMDU as scaffolding and fill them with the behavioural and cognitive insights drawn from sport. The aim is to move from fragmented understanding toward a unified model of adaptive intelligence that is rigorous, explainable, and transferrable across domains.

3 Methodology

This chapter outlines the methodological approach adopted in this study. It explains the research design, case selection logic, data sources, and analytical framework used to examine decision-making under deep uncertainty through football-based case studies. The chapter also details how qualitative comparative logic, structured analogical reasoning, and cognitive mapping are integrated to enable systematic analysis and conceptual transfer. Issues of validity, traceability, and methodological limitations are addressed to clarify the scope and rigor of the research.

3.1 Research Design

This research adopts a Qualitative Comparative Case Analysis (QCA) framework integrated with structured analogical reasoning. The aim is to examine how decision-making under deep uncertainty can be understood through football's dynamic context and then transferred into business domains. The combination of qualitative granularity and comparative logic allows for both cognitive depth and analytical consistency. Rather than seeking statistical generalization, the study prioritizes conceptual transfer and interpretive accuracy, aligning with the methodological stance of DMDU.[74]

3.1.1 Purpose and Orientation

The study is designed to explore how structured decision frameworks such as Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT) can intersect with cognitive and behavioural processes observed in football. Traditional DMDU research has focused on large-scale policy domains such as climate planning, transport, or infrastructure. However, this project extends those methods to human-scale decision environments where pressure, emotion, and intuition play crucial roles. By analysing high-stakes football matches, the research isolates observable decision instances that mirror uncertainty in corporate or institutional contexts.

The rationale for a qualitative-comparative approach is rooted in the cognitive nature of the research questions. Decision-making under uncertainty is not merely a computational

[74] See *Marchau et al.*, Decision Making under Deep Uncertainty, pp. 1-16.

process but a psychological one shaped by interpretation, time pressure, and experience.[75] QCA allows the study to compare cases based on the configuration of conditions rather than single variables. In this context, each football match represents a unique configuration of environmental volatility, tactical strategy, and psychological pressure. The comparative structure enables identification of causal patterns without losing the richness of qualitative context.

3.1.2 Integration of Analogical Reasoning

The second methodological layer is structured analogical reasoning, derived from the philosophical and cognitive work on analogy.[76] Analogical reasoning permits knowledge transfer from one domain (football) to another (business) through systematic mapping of relations rather than superficial similarities. This approach justifies the thesis' central design choice: using football as a cognitive simulation of organizational decision-making under uncertainty. The mapping structure follows three levels:

1. Source structure: football match dynamics, decision points, tactical adjustments.
2. Cognitive structure: recognition patterns, timing, and adaptation mechanisms.
3. Target structure: business contexts such as crisis management or strategic planning.

This three-level mapping ensures that each football event or managerial quote corresponds to an equivalent process in decision science, preserving relational fidelity between domains.

3.1.3 Why Qualitative Granularity Is Necessary

Quantitative performance data in sport capture what happened but not how decisions emerged. Understanding adaptive intelligence requires attention to timing, context, and cognition. Qualitative granularity makes it possible to trace the decision logic embedded in each event or statement. It also exposes the reasoning process rather than only its outcomes. This approach aligns with DMDU's emphasis on deliberation with analysis, where interpretive engagement complements model-based exploration.[77]

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[75] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 146-147.

[76] See Bartha, P., Analogical Reasoning, 2019, p. 4.

[77] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 387-389.

In practical terms, qualitative granularity allows for the coding of cognitive indicators such as recognition cues, shifts in attention, and coordination between decision agents. Football provides a controlled yet natural laboratory for observing these variables in real time. Unlike laboratory experiments, matches produce organic decision data under genuine uncertainty. The qualitative-comparative framework therefore captures both structured and emergent properties of adaptive decision-making.

3.1.4 Case Structure and Triangulation

The research uses twenty high-stakes football matches as primary cases. Each case is analysed through four dimensions derived from the coding framework: situational uncertainty, tactical adaptation, communication response, and outcome stability. These dimensions correspond to the analytical logic of RDM and DAPP, which identify vulnerabilities, triggers, and adaptive responses.[78] In parallel, five semi-structured interviews with professionals in finance, analytics, and consulting validate the transferability of insights from football to business. The interviews serve as analogical anchors, confirming whether identified decision dynamics resonate with organizational practice.

Triangulation occurs at three levels:

- Data triangulation: football event data, interview narratives, and coded decision outcomes.
- Method triangulation: qualitative content analysis, case comparison, and analogical reasoning.
- Theory triangulation: linking DMDU concepts of robustness and adaptability with cognitive decision models like OODA and RPD.

This layered structure ensures methodological rigor while maintaining interpretive flexibility.

3.1.5 Cognitive Mapping as Analytical Output

The analytical objective is to construct a cognitive decision map that links structured uncertainty frameworks to real-time decision behaviour. The cognitive map integrates

[78] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 54-56

theoretical constructs such as signposts, tipping points, and feedback loops from DMDU with behavioural elements identified in football and interview data. By aligning these levels, the research generates a framework that reflects both the structure and fluidity of human decision-making.

The map does not aim for prediction but for explanation. It visualizes how decisions evolve through feedback, recognition, and adaptation, the same principles that DMDU embeds at a policy level but now transposed to human cognition. This methodological design therefore positions qualitative insight as the bridge between structured decision models and real-world adaptability.

3.1.6 Summary

The overall design combines comparative rigor with interpretive depth. The use of QCA ensures that patterns of adaptation can be systematically identified across multiple football cases, while analogical reasoning provides the logical bridge to business environments. Qualitative granularity allows the research to reconstruct cognitive mechanisms that cannot be observed through numerical data alone. By grounding these insights within the DMDU paradigm, the study establishes a robust yet human-centred methodology for exploring decision-making under uncertainty.

3.2 Case Selection Criteria

The selection of cases in this study followed a deliberate, multi-dimensional process to capture decision-making under high uncertainty in football's most intense competitive environments. Twenty matches were chosen between 2016 and 2023, covering tournaments such as the FIFA World Cup, UEFA Champions League, and UEFA European Championship. Each match represents a real-time laboratory of decision stress, containing turning points, psychological strain, and managerial adaptation that reflect similar conditions to high-stakes business decisions.

The case selection process was guided by five linked criteria: contextual uncertainty, strategic significance, temporal diversity, managerial contrast, and analytical richness.

3.2.1 Contextual Uncertainty

All selected matches involved environments of deep uncertainty where outcomes could not be reliably forecasted. This aligns with the DMDU framework, which defines uncertainty as a state where models, probabilities, and preferences cannot be confidently established.[79] The chosen games feature volatile conditions such as red cards, comeback scenarios, penalty shootouts, and tactical gambles. These represent moments when information is incomplete, time is limited, and decision-makers must rely on adaptive reasoning rather than prediction.

3.2.2 Strategic Significance

Every match was drawn from knockout or decisive stages, where a single outcome determined survival or elimination. This structure amplifies uncertainty by removing the opportunity for gradual correction. The Argentina versus France 2022 World Cup Final and Barcelona versus PSG 2017 Champions League Round of 16 are examples of matches where momentum shifted drastically and outcomes reversed within minutes. These cases parallel strategic turning points in business, where leaders must make rapid decisions under stress and limited visibility.

3.2.3 Temporal and Geographic Diversity

The selected cases span nine years and multiple regions to capture a range of competitive contexts. By combining data from European club competitions and global tournaments, the study captures variation in tactical culture and decision-making styles. National teams and club environments present different types of uncertainty, which broadens the scope of comparison. This diversity reflects DMDU's emphasis on context-sensitive and adaptive frameworks.[80]

3.2.4 Managerial and Philosophical Contrast

A key selection principle was diversity in managerial philosophy. The sample includes reactive coaches like Didier Deschamps and José Mourinho, and proactive strategists such as Pep Guardiola and Jürgen Klopp. Bringing both groups together makes the comparison stronger because you can see how different decision logics respond to the same kind of pressure. Some managers depend on strict control, fixed structures and predefined roles, while others build systems around fluid movement, adaptability and rapid adjustment. This spread creates a better foundation for structured analogical

[79] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 1-17.

[80] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 379-382.

reasoning, since the approach works best when the underlying decision patterns vary in meaning and intent rather than just appearance.[81]

3.2.5 Analytical Richness and Traceability

Only matches with complete and verifiable event data were included. These datasets were obtained from Hudl, Wyscout, allowing consistent access to match PDFs, event-level statistics, and tactical sequences. This ensured that analytical interpretations could be traced back to source data and cross-verified with interview insights. The transparency of event coding supports reproducibility and reinforces the study's methodological integrity, consistent with DAPP's structured monitoring and signpost logic.[82]

The final selection was organized into four analytical clusters:

1. Crisis and Recovery
2. Tactical Adaptation
3. Leadership Under Pressure
4. Controversy and Unpredictability

Each cluster groups five matches that share similar types of uncertainty and decision characteristics. Together, they form a balanced dataset for comparative analysis, allowing cross-case insights while preserving contextual richness. These clusters will serve as the foundation for the analytical chapters that follow.

3.3 Data Collection

This study integrates two complementary data sources to capture how decision-making unfolds under uncertainty: (1) detailed football match reports that reflect high-pressure, real-time tactical decisions, and (2) semi-structured interviews with professionals in business contexts who navigate similar cognitive and organizational constraints. The combination provides both the structural and human dimensions of decision-making under uncertainty.

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[81] See *Lee, A.Y.*, Analogical Reasoning: A New Look at an Old Problem, 1990, pp. 1-3.

[82] See *Ben-Haim*, Info-Gap Decision Theory, 2019, pp. 957-99

The first dataset consists of twenty elite football matches from 2016 to 2023, selected from tournaments characterised by high stakes and unpredictability such as the FIFA World Cup and the UEFA Champions League. Each match was obtained in PDF format from scouting and performance analysis platforms like Wyscout and HUDL. These reports contain structured descriptions of key events, tactical evolutions, and contextual information about formations, substitutions, goals, disciplinary incidents, and referee interventions. Since these PDFs are narrative rather than numeric, they were treated as qualitative case material rather than statistical input. Each file was manually reviewed to extract specific segments that illustrate changing game dynamics, such as substitutions that altered momentum, goals that redefined team strategies, or disciplinary decisions that forced tactical recalibration.

For every match, two outputs were created. A short analytic note interpreting how each event potentially shifted decision pathways for the manager or team. These notes do not involve coding but serve as structured case summaries that identify where uncertainty became visible in the match flow. This approach follows qualitative case comparison principles within decision science, where contextual traces of uncertainty are treated as decision cues rather than numerical variables.[83]

The second dataset consists of semi-structured interviews with professionals in financial services and technology, who work in environments characterised by ambiguity, volatility, and incomplete information. The interview design aimed to mirror the five research questions of this study. Questions explored how individuals recognise relevant signals, prioritise competing information, respond to unexpected changes, and balance intuition with structured frameworks. Participants were encouraged to describe concrete decision episodes, particularly those involving time pressure or conflicting objectives.

Each interview lasted between 30 and 60 minutes and was recorded or transcribed from detailed notes depending on consent. The transcripts were then cleaned to correct transcription errors due to varying accents and noise. After transcription, interviews were coded using a rule-based qualitative content analysis method.[84]

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[83] See *Marchau, V. et al.*, Dynamic Adaptive Planning, 2019, pp. 169-171.

[84] See *Kuckartz, U.*, Qualitative Content Analysis, 2023, pp. 45-48.

The coding framework was structured around five core themes aligned with the research questions:

1. Information signals and decision structures
2. Experience of high-stakes choices
3. Perception of turning points
4. Reliance on frameworks versus intuition
5. Opportunities for decision-support systems

Unlike the football data, which provides contextual grounding, the interview data captures cognitive processes and reasoning structures. Together they create a dual-layered evidence base: football matches illustrate decision environments where uncertainty is explicit and time-bound, while interviews reveal how decision-makers conceptualise and rationalise similar pressures in organisational contexts. The two sources are therefore connected through analogy and triangulation rather than statistical aggregation.

Data integration occurs through comparative mapping. After each interview was coded, themes were aligned with similar decision moments observed in the football match notes. For instance, interviewees' reflections on transparency and adaptive timing were compared to moments in matches where managers made substitutions or formation changes following goals or red cards. This triangulation approach supports internal validity by grounding abstract reflections in concrete, observable events.

Reliability was maintained through consistent procedures in data extraction and coding. For football data, the same event categories were applied to every match. For interviews, coding consistency was tested by revisiting selected transcripts to confirm that similar phrases or decision logics were classified under the same category. To ensure transparency, an audit trail was kept linking each coded segment or event note to its original source file.

By combining structured football case data with cognitive interview evidence, this study aligns with the DMDU tradition of integrating exploratory modelling and adaptive reasoning within qualitative frameworks.[85] The approach avoids overreliance on

[85] See *Kwakkel, J.H.; Haasnoot, M.*, Supporting DMDU, 2019, pp. 357-359.

probabilistic prediction and instead focuses on identifying how individuals and systems remain adaptive, resilient, and responsive when faced with unfolding uncertainty.

3.4 Analytical Framework

The analytical framework used in this study connects structured decision-science models with the qualitative interpretation of uncertainty in football and business contexts. Its purpose is to examine how decisions evolve when time is limited, information is incomplete, and outcomes are uncertain. The framework draws from Decision Making under Deep Uncertainty (DMDU) theory and interprets football matches as dynamic, observable cases of human decision behaviour. The interviews with business professionals complement this by revealing the reasoning processes that guide similar choices in organizational settings.

At the core of the analysis is a three-layer mapping logic. The first layer is the event layer, which represents football matches as sequences of tactical decisions under pressure. The second is the cognitive layer, built from interview data that capture how professionals perceive, prioritise, and adapt to uncertainty. The third is the theoretical layer, which provides the interpretive structure through established approaches such as Robust Decision Making (RDM), Dynamic Adaptive Policy Pathways (DAPP), and Info-Gap Decision Theory (IGDT).[86], [87], [88]

Each football match is treated as a micro-simulation of decision dynamics. Every tactical event such as substitutions, goals, and formation changes, is viewed as a decision point made under pressure. This parallels how executives react to abrupt market or policy shifts. Football provides a bounded environment in which cause-and-effect can be observed with clarity, allowing patterns of adaptation to be examined in detail. These events are categorised as proactive, reactive, or adaptive, depending on the type and timing of response.

[86] See *Lempert, R. J., Robust* Decision Making, 2019, pp. 23-51.

[87] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 71-92.

[88] See *Ben-Haim*, Info-Gap Decision Theory, 2019, pp. 93-113.

The analytical process proceeds through four interpretive stages:

Stage 1: Event Identification

Each match is reviewed to identify situations that represent uncertainty, such as disciplinary incidents, tactical shifts, or time-critical substitutions. Events are described by their timing and context. For example, a late substitution after conceding a goal is coded as a reactive decision, while a halftime change anticipating fatigue is coded as proactive.

Stage 2: Cognitive Alignment

The second stage aligns these football events with insights from the interview data. When a manager describes acting with incomplete information, it parallels a coach making tactical adjustments without full visibility of the opponent's intentions. The interviews therefore ground the football cases within authentic cognitive reasoning patterns, supporting what Marchau, Walker, and Bloemen describe as "deliberation with analysis".[89]

Stage 3: Theoretical Integration

This stage links the empirical material to formal decision-science frameworks. RDM explains how actors select strategies that perform satisfactorily across a range of possible futures rather than optimising for a single forecast.[90] DAPP contributes the notion of adaptation tipping points, moments where a current strategy loses effectiveness and must evolve.[91] In football, these translate into tactical flexibility, resilience to setbacks, and recognition of when a plan must change.

Stage 4: Cross-Domain Comparison

The final stage compares decision behaviour across football and business. Drawing on established work in analogical reasoning,[92] the analysis treats football as an interpretive analogue rather than a model to replicate. A coach deciding whether to defend or press higher resembles an executive choosing whether to consolidate or expand amid

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[89] See *Marchau et al*., Decision Making under Deep Uncertainty, p. 16.

[90] See *Lempert, R. J., Robust* Decision Making, 2019, pp. 23-25.

[91] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 71-74.

[92] See *Holyoak, K.J.; Morrison, R.G.*, The Oxford Handbook of Thinking and Reasoning, 2012, pp. 3-5.

uncertainty. This comparative reasoning highlights structural similarities in adaptive cognition while preserving contextual differences.

The outcome of these stages is the Decision Intelligence Transfer Framework (DITF). The DITF operates along three analytical axes: situational awareness, adaptive action, and systemic consequence.

This framework also builds on Kay and King's argument that uncertainty cannot be captured through probabilistic precision but must be understood through narrative reasoning and adaptive interpretation.[93] In this study, the same principle applies to both tactical and organizational contexts: decision quality depends less on predictive accuracy and more on flexibility, perception, and judgment.

By combining football case analysis with DMDU principles and the cognitive evidence drawn from interviews, this framework presents football not as a metaphor to be copied but as a structured comparative system. It enables uncertainty, adaptation, and resilience to be observed and interpreted in parallel. The DITF therefore positions football decision-making as a lens through which organizational behaviour can be examined, revealing how effective decision intelligence arises from reasoning under pressure rather than deterministic planning.

3.5 Validity and Limitations

The validity of this research depends on how accurately qualitative interpretation captures decision-making under uncertainty without imposing artificial structure or hindsight bias. Because the study combines football case analysis, interview data, and established decision-science frameworks, its reliability is grounded in triangulation rather than statistical validation. The football data, derived from Wyscout match reports, provide an objective record of events, while the interviews introduce context-rich perspectives that show how decision-makers perceive and act within uncertain environments.

The qualitative nature of this material introduces subjectivity. Classifying events in football matches such as whether a substitution represents proactive planning or reactive adjustment relies on interpretation rather than fixed metrics. To reduce bias, football events are cross-referenced with coded interview insights to ensure consistency in how

[93] See *Kay, J.; King, M.*, Radical Uncertainty, 2020, pp. 48-49.

uncertainty is understood across both datasets. As Lempert argues in his discussion of Robust Decision Making, even structured models depend on framing assumptions that influence the interpretation of robustness.[94]

A further limitation comes from the analogical method itself. Translating insights from football into business carries the risk of analogical bias when surface similarities are mistaken for deeper structural parallels. This study follows the disciplined mapping approach described by Holyoak.[95] The validity of such comparisons depends on distinguishing between analogy and equivalence. The analysis therefore focuses on decision patterns such as adaptation, recovery, or escalation control rather than outcomes or performance measures.

Context dependence is another limitation. Football events occur under extreme emotional and institutional pressure within a competitive setting that does not perfectly mirror corporate or policy environments. As Kay and King explain in Radical Uncertainty: Decision-Making Beyond the Numbers, uncertainty in human systems cannot be reduced to probability because it is embedded in social expectations and narrative framing.[96] Football is therefore treated as a structured observation of human adaptation rather than a literal representation of business decision-making.

The limited scale of this study also restricts generalization. The analysis draws on twenty high-stakes football matches and five semi-structured interviews. These sources provide qualitative richness but not statistical universality. The findings are analytically generalizable, extending theoretical understanding of how adaptive intelligence functions under deep uncertainty rather than producing predictive laws.

Overall, the research maintains validity through conceptual transparency and consistency. The integration of football analysis, cognitive interpretation, and theoretical grounding creates a coherent framework that accepts its limits. Instead of aiming for prediction, it demonstrates how structured decision models can align with real-world adaptability and human reasoning.

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[94] See *Lempert, R.J.*, Robust Decision Making, 2019, pp. 27-30.

[95] See *Keith J. Holyoak*, “Analogy and Relational Reasoning,” pp. 234-259.

[96] See *Kay, J.; King, M.*, Radical Uncertainty, 2020, pp. 48-49.

3.6 Interview Analysis and Coding

The qualitative interview component was designed to provide a human-centred perspective on decision-making under uncertainty, complementing the structured football match data used in this study. While match events offer a controlled environment to observe tactical and time-pressured choices, interviews reveal how experienced professionals interpret, rationalize, and adapt their decisions when faced with uncertainty in real-world contexts. The aim was not only to triangulate the football-derived observations but also to understand how individuals in business and data-driven domains internalize ambiguity, manage pressure, and structure decisions when time and information are limited.

Five professionals were selected through purposive sampling of which two of them opted to remain anonymous in the thesis and only allowed to use their position and company to refer them, they both will be referred to as “Interviewee A’ and “Interviewee B” in the Thesis.

These professionals were selected to ensure relevance to the study’s conceptual framework. All participants held decision-making responsibilities in high-stakes or data-dependent environments. Daniel Jarzynski, Senior Manager at HSBC ALCM in Poland, provided insight into risk and liquidity management within financial systems. Interviewee A, Senior Manager at HSBC ALCM Germany, and Interviewee B, a working student at the same department, both offered perspectives on regulatory reporting, liquidity forecasting, and the operational side of uncertainty management while this also highlighted the difference in decision making based on experience. Adnan Mushtaq, Data Scientist at SURU, Germany, contributed a technological viewpoint on algorithmic decision-making and adaptive modelling. Finally, Manzar Shaikh, Assistant Vice President and Product Lead at FatakPay, India, provided the entrepreneurial and product innovation perspective, where uncertainty often intersects with market shifts and customer behaviour.

All interviews were conducted online between August and October 2025, lasting between 35 and 60 minutes. Participants were informed of the study’s purpose, data confidentiality, and their right to withdraw at any stage. Consent was obtained through a consent form as well as verbally and recorded before each session. Ethical considerations were handled in accordance with standard qualitative research procedures.[97]

Each interview followed a semi-structured format aligned with the five research questions defined in Chapter 1. The structure ensured comparability while allowing participants to elaborate freely on individual experiences. Thematic prompts were organized as follows: (1) signals and structure in decision-making under uncertainty, (2) experiences with high-stakes decision environments, (3) perception of turning points and critical events, (4) frameworks or heuristics used when facing uncertainty, and (5) reflections on decision-support systems. This design mirrored the five research questions (RQ1 - RQ5) of the thesis and created a clear mapping between theoretical constructs and practical experience.

The interviews were transcribed verbatim and analysed using a deductive-inductive hybrid coding approach. A set of pre-defined codes derived from the research questions guided initial categorization (e.g., “pattern recognition,” “time compression,” “adaptive framing”), while emergent sub-codes captured variation within each participant’s reasoning. Each code was then grouped into broader analytical clusters, corresponding to key thematic areas of the thesis, Crisis and Recovery, Momentum and Adaptation, Strategy Calibration, and Cognitive Framing. This multi-layered process allowed the interview data to serve not just as anecdotal support but as analytical evidence integrated with the football case studies.

For example, within the Crisis and Recovery cluster, Daniel Jarzynski described how “rapid restructuring during financial volatility mirrors tactical substitutions when momentum shifts unexpectedly.”[98] Similarly, Interviewee A emphasized that “the credibility of incoming data matters more than its completeness” when under operational pressure.[99] In the Strategy Calibration cluster, Adnan Mushtaq compared algorithmic re-tuning in data pipelines to halftime adjustments in football, noting that “models and managers both require structured feedback loops to stay relevant.”[100] These parallels

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[97] See *Flick, U.*, The SAGE Handbook of Qualitative Research, 2014, pp. 47-50.

[98] *Jarzynski D.*, interview conducted on 08.09.2025.

[99] *Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.*

[100] *Mushtaq A.*, interview conducted on 10.10.2025.

illustrate the interpretive bridge between structured decision frameworks and experiential adaptability, the core premise of this thesis.

The coding matrix (see Appendix A) organizes these findings across two dimensions:

1. Cognitive focus: how individuals perceive and process uncertainty (pattern recognition, intuition, rule adherence, and emotional regulation).
2. Operational response: how they structure actions, delegate authority, and adjust strategies under stress.

This structure mirrors the logic of Robust Decision-Making and Adaptive Policy Pathways, where strategies evolve dynamically based on feedback and contextual signals.[101], [102]

The interviews also revealed consistent cognitive themes. Participants described three dominant modes of reasoning:

1. Intuitive synthesis, drawing from experience to make quick, pattern-based judgments.
2. Structured adaptivity, using mental checklists or trigger points that mirror adaptive planning under uncertainty
3. Collaborative sense-making, managing uncertainty collectively through dialogue and distributed decision authority.

While a couple of participants operated in different industries, a unifying observation emerged: decisions perceived as successful were those that displayed flexibility and legitimacy rather than simply achieving speed. This aligns with the principle that robust strategies aim to remain viable across changing conditions rather than optimizing for one outcome.[103]

In summary, the interview analysis provides a qualitative backbone for interpreting the football case studies. It translates abstract decision frameworks into lived cognition and professional reflexes. By juxtaposing match-based tactical events with human reasoning

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[101] See *Lempert, R. J., Robust* Decision Making, 2019, p. 38.

[102] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, p. 72.

[103] See *Lempert, R. J., Robust* Decision Making, 2019, p. 30.

patterns, this thesis situates decision-making under uncertainty as both a structured science and a human practice. The interviews thus serve not merely as supplementary material but as interpretive scaffolding through which adaptive intelligence, the synthesis of analytical models and human adaptability can be understood in its full cognitive and organizational depth.

4 Analysis and Case Studies

The analytical phase of this thesis explores how decision-making under uncertainty manifests across real football events and mirrors organizational responses in high-stakes business environments. Each selected match functions as a qualitative simulation that captures the dynamics of crisis, adaptation, leadership, and unpredictability within structured yet volatile systems. The objective is not to predict outcomes but to understand how agents/managers/coaching staff think, act, and recover when confronted with deep uncertainty. This approach draws directly from the Decision Making under Deep Uncertainty (DMDU) literature, which emphasizes robustness, flexibility, and learning over optimization.[104]

To analyse these dynamics, four thematic clusters were derived from the coded interview data and mapped onto selected matches from 2016 to 2023. Each cluster represents a dimension of adaptive behaviour:

1. Crisis and Recovery, examining how teams and leaders regain control after setbacks.
2. Tactical Adaptation, focusing on agility and responsiveness to real-time feedback.
3. Leadership Under Pressure, assessing decision quality under stress, time compression, and collective dependence.
4. Controversy and Unpredictability, exploring how fairness, regulation, and moral uncertainty shape outcomes.

These clusters emerged from the interview coding process, which identified recurring patterns such as situational awareness, timing of intervention, and mental resilience. Daniel Jarzynski described this as “recognizing the difference between reacting fast and reacting correctly,” a distinction that underlines the essence of decision quality in volatile conditions.[105] Similarly, Adnan Mushtaq observed that “data can guide the frame, but

[104] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 15.

[105] *Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.*

intuition defines the move," emphasizing that real-time cognition is a negotiation between information and instinct.[106]

From a methodological perspective, each cluster integrates two analytical layers:

1. Empirical observation, using event-level match data extracted from Wyscout PDFs (goals, substitutions, disciplinary events, formation changes).
2. Interpretive synthesis, grounded in DMDU principles and qualitative reasoning from the interview corpus.

This dual approach allows for linking structured frameworks, such as Robust Decision Making, with the cognitive models found in sports decision-making research, including Klein's Recognition-Primed Decision framework[107].

The resulting synthesis is interpretive rather than statistical, positioning football as an experimental ground for decision intelligence. It allows the observation of how structured models (RDM, DAPP, IGDT) manifest in unpredictable human environments and how lessons can transfer to business settings where uncertainty is constant, and stakes are high.

The following sections analyse each cluster through five mini-cases, using literature and interview triangulation to trace how coaches, players, or managers construct meaning, recover from setbacks, and adapt under constraints. The aim is to reveal decision logic that is not linear but iterative, embedded in context, and shaped by perception, evaluation, and adaptation.

4.1 Cluster 1 - Crisis and Recovery

Periods of crisis often reveal more about the quality of decision-making than moments of control. In both football and business, decisions made under pressure determine not only immediate survival but long-term adaptability. This cluster examines five matches that embody rapid recovery from disruption, using them to explore the mechanisms of crisis response, learning, and turnaround strategy. The cases illustrate how robust and adaptive decision frameworks, such as those proposed in Decision Making under Deep Uncertainty (DMDU) approaches like Robust Decision Making (RDM) and Dynamic Adaptive Policy

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[106] *Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.*

[107] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 138-147.

Pathways (DAPP), can be observed in action within real-world tactical contexts. Each match is treated as a micro-simulation of how human and structural intelligence interact under uncertainty.[108]

Case 1: Argentina vs France (FIFA World Cup Final, 2022)

The 2022 World Cup Final represented a confrontation between two systems of adaptive intelligence. Argentina's early dominance was disrupted by France's sudden resurgence in the final twenty minutes, driven by Mbappé's rapid double strike.[109] What followed was a case study in resilience and iterative recalibration. Scaloni's response was not based on complex probabilistic reasoning but on the recognition of momentum signals and emotional state management. From a DMDU perspective, this aligns with the principle of "adaptive signposts" described by Haasnoot et al.[110]

In interview discussions, Daniel Jarzynski described a similar process in financial stress testing where he explained that in liquidity scenarios, when a shock hits, we do not wait for full data to arrive. We start with directional indicators and adjust as the situation unfolds.[111] This mirrors Scaloni's tactical adjustment after France's equalizer, where the substitution of Paredes stabilized midfield tempo before the penalty shootout.[112] The match underlines how decision quality under crisis depends on fast interpretation of incomplete data and controlled recalibration rather than assuming and planning.

Case 2: Belgium vs Japan (FIFA World Cup Round of 16, 2018)

This match remains one of the clearest demonstrations of structured adaptability under uncertainty. Trailing 0-2 with thirty minutes remaining, Belgium faced an environment of collapsing probability. Rather than abandoning system coherence, the team applied what could be termed a "Dynamic Adaptive Plan," substituting Fellaini and Chadli to alter physical structure and passing geometry.[113] Within twenty-five minutes, the deficit

[108] See *Bonjean Stanton, M.C.; Roelich, K.*, Decision Making under Deep Uncertainty, 2021, pp. 1-4.

[109] See *Wyscout*, Match Report: Argentina vs France, FIFA World Cup Final, 2022.

[110] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 72-74.

[111] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.

[112] See *Wyscout*, Full Match Video: Argentina vs France, FIFA World Cup Final, 2022

[113] See *Wyscout*, Match Report: Belgium vs Japan, FIFA World Cup Round of 16, 2018.

turned into a 3-2 victory, culminating in a 94th-minute counterattack initiated by Courtois and De Bruyne.[114]

In business terms, this represents a transition from defensive containment to offensive opportunity recognition. Adnan Mushtaq, reflecting on AI deployment under project failures, noted that when a model starts collapsing, the first instinct should not be to discard it but to identify which variables are still working. Crisis gives you feedback loops in real time.[115] His comment directly echoes Boyd's OODA loop concept, where continuous observation and reorientation guide effective responses.[116]

The Belgium comeback illustrates RDM in practice. Instead of predicting precise outcomes, decisions were tested for robustness under extreme variability. The coach's substitutions effectively stress-tested multiple tactical pathways simultaneously, reducing vulnerability to residual uncertainty. This case shows how resilience emerges from feedback-aware leadership rather than rigid adherence to initial plans.[117]

Case 3: Croatia vs England (FIFA World Cup Semi-final, 2018)

Croatia's extra-time victory over England encapsulated the importance of cumulative resilience in uncertain systems. After conceding an early goal, Croatia maintained structural discipline rather than reacting impulsively.[118] The turning point came in the second half when Modrić's midfield rotations began to absorb England's press. The decision to sustain possession despite fatigue reflected confidence in long-term equilibrium rather than short-term disruption management.[119]

Manzar Shaikh drew a comparable analogy in the product domain, when pressure mounts, the hardest part is to avoid premature reaction. Sometimes the right choice is to hold your structure, not to over-correct.[120] His reflection aligns with Klein's Recognition-Primed

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[114] See *Wyscout*, Full Match Video: Belgium vs Japan, FIFA World Cup Round of 16, 2018.

[115] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

[116] See *Horney, N.*, OODA Loop, 2011, pp. 2-4.

[117] See *Lempert, R. J., Robust* Decision Making, 2019, pp. 30-32.

[118] See *Wyscout*, Match Report: Croatia vs England, FIFA World Cup Semi-Finals, 2018.

[119] See *Wyscout*, Full Match Video: Croatia vs England, FIFA World Cup Semi-Finals, 2018.

[120] Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.

Decision model, where experienced actors simulate outcomes mentally before committing to change.[121]

From a theoretical perspective, Croatia's approach resonates with Info-Gap Decision Theory.[122] By operating under acknowledged uncertainty, the team minimized vulnerability to worst-case scenarios. Modrić's composure in reorienting play through Brozović reflected a search for opportunity within bounded knowledge. The win demonstrated that resilience in uncertain environments is less about control and more about managing the space between predictability and chaos.

Case 4: Liverpool vs Dortmund (UEFA Europa League Quarter-Final, 2016)

The quarterfinal second leg between Liverpool and Borussia Dortmund is remembered as one of the most dramatic and emotional comebacks in modern football.[123] Facing a 1-3 aggregate deficit, Liverpool's transformation in the second half represented the mobilization of collective belief as a decision asset. Klopp's halftime adjustment pushing full-backs higher and allowing Coutinho to roam inside was not a pre-scripted decision but an emergent adaptation to energy shifts on the field.[124]

Interviewee A, who manages liquidity stress indicators at HSBC, made an interesting observation: "When models start breaking apart, our best response often comes from the people closest to the data. Top-down commands fail in that chaos."[125] His insight parallels Klopp's on-field decentralization of responsibility. Players like Milner and Can were given freedom to interpret transitions autonomously, echoing the distributed adaptation mechanisms central to Engineering Options Analysis.[126]

The comeback illustrates the principle that resilience is not only structural but emotional. As Stirling argues, robustness without diversity often collapses under stress, whereas

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[121] See *Klein. G.*, Recognition-Primed Decision Model, 1993, p. 145.

[122] See Ben-Haim, Info-Gap Decision Theory, 2019, pp. 106-108.

[123] See *BBC Sport*, "Liverpool 4-3 Borussia Dortmund (agg: 5-4)," 14 April 2016, no page number.

[124] See *The Guardian, Liverpool v Borussia Dortmund,* Europa League Quarter-Final, 2016, no page number.

[125] *Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.*

[126] See *Neufville, R.; Smet, K.*, Engineering Options Analysis, 2019, pp. 126-127.

diversity of perspectives generates recoverability.[127] Liverpool's diversity in tactical roles and individual expression allowed the system to remain viable under turbulence.

Case 5: Roma vs Barcelona (UEFA Champions League Quarter-Final, 2018)

Roma's 3-0 victory over Barcelona after a 1-4 first leg loss was a masterclass in disciplined risk-taking under extreme uncertainty. The tactical strategy was designed not to predict the exact conditions for success but to maintain a range of options robust enough to exploit any instability in Barcelona's defence.[128] The decision to man-mark Busquets and push full-backs aggressively was consistent with the RDM logic of exploring "safe-to-fail" experiments.

Interviewee B, reflecting on liquidity planning models, remarked that sometimes, a plan's strength lies not in precision but in how quickly it can be rebuilt.[129] His view connects with Walker et al.,[130] who emphasize adaptability as a strategy for evolving uncertainty.

Roma's sustained pressure was supported by constant recalibration of tempo, a form of collective feedback adaptation. Like a business recovering from structural shock, the team did not attempt to eliminate uncertainty but to harness it. Their third goal in the 82nd minute symbolized a full system rebound.[131]

4.1.1 Synthesis of Cluster 1

Across these five cases, crisis management emerges as a process of structured improvisation rather than rigid control. Each event demonstrates a balance between planning and adaptation consistent with DMDU principles. The interviews reinforce that in both football and business; high performers do not eliminate uncertainty but operate effectively within it. The shared thread across all examples is the capacity to extract meaning from disorder, transforming volatility into feedback and recovery into learning.

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[127] See *Andy Stirling*, "Keep it Complex," Nature, Vol. 468, 23/30 December 2010, pp. 1029-1031.

[128] See *Holding Midfield*, "Case Study: AS Roma 3-0 Barcelona (10/4/2018)," 10 April 2018, no page number.

[129] Interview with Interviewee B, conducted on 11.09.2025. See Appendix B.2.2.

[130] See *Marchau et al.*, Decision Making under Deep Uncertainty, pp. 1-17.

[131] See *Wyscout*, Full Match Video: Roma vs Barcelona, UEFA Champions League Quarter Finals, 2018.

4.2 Cluster 2 - Tactical Adaptation

Tactical adaptation represents the ability to make structured changes under pressure without losing the overall identity of the system. In football, this means altering formations, adjusting tempo, or redistributing creative roles while preserving the strategic goal. In decision science, this reflects the concept of adaptive robustness found in frameworks like Dynamic Adaptive Policy Pathways (DAPP) and Robust Decision-Making (RDM), where success depends on flexibility under uncertainty rather than precision under control. The following five matches illustrate different dimensions of tactical adaptation and demonstrate how principles from football strategy align with organizational agility, cognitive learning, and real-time analytical decision-making.

Case 1: Barcelona vs PSG (UEFA Champions League Round of 16, 2017)

Barcelona's comeback against PSG remains one of the most striking examples of tactical adaptation in modern football.[132] Entering the second leg with a 0-4 deficit, Luis Enrique replaced the standard 4-3-3 with a flexible 3-4-3, enabling overloads on both flanks.[133] The wide progression of Neymar and Sergi Roberto created continuous positional stress for PSG, forcing defensive instability and allowing Barcelona to dominate the tempo. [134]This transformation reflects the DMDU notion of a policy stress test, where structural changes are intentionally deployed to observe system response.[135]

In the interviews, Daniel mentioned that in crisis analysis, certainty does not exist in a running process. You act, you observe, you reweight.[136] His approach mirrors Enrique's decision-making during the match, where immediate learning replaced long planning. Adnan described a similar mechanism in model calibration: "We modify data weight in flight, even if it breaks the model temporarily."[137] Both comments align with how

[132] See *Sky Sports*, "Barcelona's epic comeback against PSG underlines their turnaround," 9 March 2017, no page number.

[133] See *Wyscout*, Match Report: Barcelona vs PSG, UEFA Champions League Round of 16, 2017.

[134] See *Wyscout*, Full Match Video: Barcelona vs PSG, UEFA Champions League Round of 16, 2017.

[135] See *Lempert, R. J.,* Robust Decision Making, 2019, p. 25.

[136] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.

[137] *Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.*

Barcelona's risk-heavy structure operated, showing that effective adaptation under pressure relies on iterative recalibration rather than static control.

Manzar added that conviction sometimes has to precede clarity, and leaders must decide before they fully understand.[138] This parallels Enrique's belief-driven substitutions in the last 10 minutes when Neymar took over creative command.[139] In DMDU terms, this reflects an adaptive commitment under deep uncertainty, where resilience is built through active response rather than passive waiting.[140]

Case 2: France vs Australia (FIFA World Cup Group Stage, 2018)

France's group-stage match against Australia demonstrates a more structured form of adaptation. Didier Deschamps introduced Antoine Griezmann in a deeper role to link midfield transitions and later replaced him with Nabil Fekir to stabilize the attack.[141] The adjustments were not radical but incremental, designed to rebalance control. This kind of slow recalibration mirrors adaptive pathways in decision science, where small course corrections preserve system stability over time.[142]

Daniel described this in financial decision-making as maintaining the core assumption but testing parameters on the periphery.[143] His view corresponds to Deschamps' model of maintaining shape while subtly changing influence zones. Adnan observed that models should degrade gracefully rather than fail suddenly.[144] This resonates with the logic of adaptive planning, where flexibility is a built-in safeguard rather than a reaction.

France's tactical evolution during the game illustrated what RDM defines as robustness through diversity. Instead of searching for an optimal setup, the team created multiple working states that could shift fluidly as the match evolved. This strategic patience aligns with the interviews' consistent theme that adaptability is not improvisation but managed variability.

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[138] Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.

[139] See *Wyscout*, Full Match Video: Barcelona vs PSG, UEFA Champions League Quarter Finals, 2017.

[140] See *Kay, J.; King, M.*, Radical Uncertainty, 2020, pp. 20-22.

[141] See *Wyscout*, Match Report: France vs Australia, FIFA World Cup Group Stage, 2018.

[142] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 72-74.

[143] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.

[144] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

Case 3: Manchester City vs Liverpool ((UEFA Champions League Quarter-Final, 2018)

This match highlighted adaptive interaction between two systems in competition. Pep Guardiola introduced Bernardo Silva in a deeper role to control possession rhythm, aiming to neutralize Liverpool's pressing. Jürgen Klopp countered by moving James Milner into central midfield, generating overloads that disrupted City's buildup.[145] Each manager's adjustment triggered a response from the other, creating a continuous feedback loop of learning.[146]

Interviewee A compared this kind of iterative interaction to internal regulatory adjustments: "If one control shifts, the rest of the system self-corrects."[147] This perspective reflects co-adaptive behaviour described in complex decision frameworks.[148] Adnan reinforced this by saying, "True adaptability happens when both sides of the system are learning at the same speed."[149]

The critical turning point was Salah's goal,[150] which inverted City's momentum. Guardiola's decision to maintain an aggressive press despite structural instability resembles the problem of model lock-in, where adherence to a known plan suppresses new feedback.[151] The interviews show a parallel: Interviewee A and Interviewee B noted that confidence can block correction. This match, therefore, serves as a micro-simulation of how adaptive failure can emerge not from lack of intelligence but from overconfidence in fixed logic.

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[145] See *Wyscout*, Match Report: Manchester City vs Liverpool, UEFA Champions League Quarter Finals, 2018.

[146] See *Wyscout*, Full Match Video: Manchester City vs Liverpool, UEFA Champions League Quarter Finals, 2018.

[147] *Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.*

[148] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 15.

[149] *Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.*

[150] See *Wyscout*, Full Match Video: Manchester City vs Liverpool, UEFA Champions League Quarter Finals, 2018.

[151] See *Taleb, N.*, The Black Swan, 2007, p. 267.

Case 4: Juventus vs Real Madrid (UEFA Champions League Semi-Final, 2015)

Juventus' victory over Real Madrid was a study in structured discipline. Massimiliano Allegri used a 4-1-4-1 shape that shifted dynamically into a 3-5-2 during transitions.[152] The system allowed Juventus to absorb Madrid's pressure and strike through vertical bursts.[153] Allegri's controlled flexibility reflects the decision-science principle of managed uncertainty, where boundaries define freedom rather than restrict it.

Daniel's interview supports this interpretation; Frameworks do not kill creativity, they structure it.[154] Interviewee B, on the other hand, noted that stability under chaos often comes from routine.[155] This insight connects directly to Juventus' rhythm in the second leg, where repetition provided predictability under stress. Adnan emphasized that structured adaptation is not reactive but anticipatory, Systems should know where they will bend before they are hit.[156]

This reflects the methodological logic of Robust Decision-Making, where adaptability is designed before the uncertainty unfolds.[157] Juventus' performance shows how stability and flexibility can coexist through disciplined learning loops.

Case 5: Iceland vs Croatia (FIFA World Cup Group Stage, 2018)

The Iceland vs Croatia match represents small-nation adaptability in the face of structural asymmetry.[158] Iceland's tactical approach shifted repeatedly from high press to mid-block, with players rotating into multiple unfamiliar roles.[159] Despite losing 1-2, Iceland displayed what in DMDU is called adaptive capacity the ability to remain effective under incomplete knowledge and limited resources.

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[152] See *Wyscout*, Match Report: Juventus vs Real Madrid, UEFA Champions League Semi Finals, 2015.

[153] See *Wyscout*, Full Match Video: Juventus vs Real Madrid, UEFA Champions League Semi Finals, 2015.

[154] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.

[155] Interview with Interviewee B, conducted on 11.09.2025. See Appendix B.2.2.

[156] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

[157] See *Lempert et al.*, Robust Decision Making, 2019, pp. 25-27.

[158] See *The Guardian*, "Iceland remain optimists despite Croatia defeat," 25 June 2018, no page number.

[159] See *Wyscout*, Match Report: Iceland vs Croatia, FIFA World Cup Group Stage, 2018.

Manzar mentioned that “sometimes being small forces clarity.”[160] This reflects Iceland’s disciplined risk-taking, showing that strategic adaptability often emerges not from power but from proportionate learning. Adnan described this as “local optimization inside global uncertainty,”[161] which corresponds directly with Iceland’s tactical responses to Croatia’s shifting formations.

This case also reinforces the interviews’ collective view that uncertainty is not only external but cognitive.[162] Daniel summarized it clearly: Decisions under stress are less about options and more about perception.[163] That perception-driven approach matches Iceland’s composure, as their players kept recalibrating positioning even when success probability was low.[164] It reflects the practical application of adaptive decision-making under bounded rationality.

4.2.1 Synthesis of Cluster 2

Across these five cases, tactical adaptation appears not as random improvisation but as structured flexibility. Managers who succeed under uncertainty do not discard frameworks but reinterpret them dynamically. The interview data consistently emphasized this principle: structured intuition, time-bound recalibration, and clarity of process even when outcomes are unpredictable.

In both football and business, tactical adaptation depends on maintaining awareness of feedback and adjusting actions proportionally rather than impulsively. The connection between Deschamps’ measured substitutions, Guardiola’s feedback loops, and Allegri’s disciplined balance reflects a common decision architecture. This cluster therefore reinforces the theoretical bridge between DMDU’s adaptive logic, and the cognitive behaviours observed in high-stakes professional settings.

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[160] *Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.*

[161] *Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.*

[162] See Appendix A: Coding Matrices

[163] *Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.*

[164] See *Wyscout*, Full Match Video: Iceland vs Croatia, FIFA World Cup Group Stage, 2018.

4.3 Cluster 3 - Leadership Under Pressure

Leadership under pressure is the purest form of decision-making exposure. The matches in this cluster capture moments where composure, clarity, and psychological control became more critical than tactical preparation. Each event involved teams pushed to their mental and physical limits, often through penalty shootouts or prolonged uncertainty. These instances mirror organizational crises where leaders must act under ambiguity, sustain team morale, and maintain clarity when structure collapses. Interview insights, particularly from Daniel and Adnan, emphasize that leadership resilience and emotional regulation form the backbone of decision confidence during such moments.

Case 1: Croatia vs Denmark (FIFA World Cup Round of 16, 2018)

This match remains one of the most revealing studies in controlled composure. Both teams scored within the first five minutes[165], and from that point on, the rhythm of the game became psychological rather than tactical. Croatia's captain Luka Modrić, despite missing a late penalty in extra time, recovered his composure to convert again during the shootout.[166] His behaviour displayed emotional resilience and collective grounding, attributes reflected in high-stress business environments where a single error cannot dictate the outcome trajectory.[167]

Daniel noted that effective leadership during uncertainty is not about suppressing emotion but channelling it toward decisive communication.[168] He said that pressure amplifies clarity for people who have rehearsed their response rather than their plan. That echoes the match's narrative, where Modrić acted as the emotional metronome, allowing his teammates to read his confidence as a stabilizing cue.

Academic studies in crisis psychology highlight this process as affective contagion, where emotional stability in one leader influences group equilibrium.[169]In both business and football, that moment of stillness before decision execution becomes the pivot between chaos and cohesion.

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[165] See *Wyscout*, Match Report: Croatia vs Denmark, FIFA World Cup Round of 16, 2018.
[166] See *Wyscout*, Full Match Video: Croatia vs Denmark, FIFA World Cup Round of 16, 2018.
[167] See *BBC Sport*, "World Cup 2018: Croatia beat Denmark on penalties to reach quarterfinals," 1 July 2018, no page number.
[168] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.
[169] See *Smith, C. A., Lazarus, R. S.*, "Emotion and Adaptation," 1990, p. 614.

Case 2: Portugal vs France (UEFA Euro Final, 2016)

Portugal's extra time 1-0 victory without Cristiano Ronaldo for most of the final displayed distributed leadership and adaptive responsibility. When Ronaldo was stretchered off, the entire psychological structure of the team shifted[170]. Yet the collective response, led by Pepe, Rui Patrício, and Eder, transformed individual loss into a shared mission.[171]

Manzar reflected on this concept by saying that leadership under uncertainty is not always about the person with the title, it's about who absorbs responsibility without being asked."[172] This mirrors how Eder, a peripheral striker, emerged as the unexpected focal point, scoring the decisive goal in extra time. The adaptive transfer of accountability among players mirrored business cases where unexpected actors assume control when formal leadership is compromised.

The match aligns with frameworks of emergent leadership where authority fluidly transfers based on situational credibility rather than hierarchy[173]. In organizations, as in this match, distributed confidence replaces centralized dependence.

Case 3: Real Madrid vs Atlético Madrid (UEFA Champions League Final, 2016)

This final remains an exemplary case of emotional regulation and experience-led leadership. Real Madrid, having faced an intense equalizer late in regular time, held psychological superiority through their captain Sergio Ramos and goalkeeper Keylor Navas.[174] The penalty shootout highlighted calm under sustained provocation, with Ronaldo sealing the win after careful sequencing of kick-takers by Zidane.[175]

Adnan emphasized during the interview that leadership under pressure is about pre-committing to calm,[176] referring to the importance of pre-event frameworks that dictate behaviour when uncertainty peaks. His statement aligns with resilience models discussed

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[170] See *The Guardian*, "Cristiano Ronaldo's tears of sadness turn to joy on Portugal's greatest night," 10 July 2016, no page number.

[171] See *BBC Sport*, "Portugal's Euro 2016 win was for Cristiano Ronaldo - Pepe," 11 July 2016, no page number.

[172] Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.

[173] See *Sowell, J.*, "A Conceptual Model of Planned Adaptation (PA)," in Decision Making under Deep Uncertainty, 2019, p. 302.

[174] See *The Guardian*, "Real Madrid win Champions League on penalties against Atlético," 28 May 2016, no page number.

[175] See Wyscout, Match Report: Real Madrid vs Atlético Madrid, UEFA Champions League Final, 2016.

[176] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

in DMDU literature, which suggest that psychological readiness can be pre-structured through cognitive rehearsal.

This balance of experience, tactical sequencing, and controlled emotion resembles managerial boards managing external shocks during market volatility.[177] The decision to allow Ronaldo to take the final penalty was less symbolic and more structurally predictive it consolidated psychological certainty in the outcome.[178]

Case 4: Netherlands vs Argentina (FIFA World Cup Quarter Final, 2022)

The 2022 World Cup quarter-final between the Netherlands and Argentina illustrated leadership under extreme emotional and structural volatility. Unlike the 2014 semi-final, this match unfolded as a psychologically unstable contest marked by escalating conflict, time-extension pressure, and disrupted control. Argentina's early dominance was challenged by late Dutch tactical adaptation, forcing decision-making under conditions of heightened uncertainty rather than pre-planned execution.

Lionel Messi's role extended beyond technical leadership into emotional regulation, acting as both a stabiliser and provocateur within a fragmented game state. In contrast, the Netherlands relied on late structural interventions rather than adaptive improvisation, highlighting the limits of delayed flexibility under pressure.

Manzar, reflecting on similar cases in project management, mentioned that over-structured preparation sometimes kills the ability to improvise.[179] This interpretation directly parallels van Gaal's situation, where the absence of flexibility became the cost of earlier control. The case mirrors the decision rigidity highlighted in adaptive systems theory, where early optimization can reduce adaptive resilience later.[180]

The psychological takeaway is clear: leadership under pressure requires timing that respects both structure and spontaneity. Decision quality deteriorates when all potential options have been predetermined.

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[177] See *Neufville, R.; Smet, K.*, Engineering Options Analysis, 2019, pp. 118-119.
[178] See Spielverlagerung, "Champions League final: Real Madrid - Atletico Madrid 1:1 (5:3 Penalties)," 30 May 2016, no page number.
[179] *Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.*
[180] See *Sowell, J.*, "A Conceptual Model of Planned Adaptation (PA)," in Decision Making under Deep Uncertainty, 2019, p. 312.

Case 5: Russia vs Croatia (FIFA World Cup Quarter-Final, 2018)

The quarter-final between Russia and Croatia extended to penalties after intense physical and emotional exhaustion.[181] Croatia's leadership, again through Modrić and Rakitić, combined emotional durability with data-guided decision patterns such as shot sequencing and recovery management.[182] What distinguished this match from the Denmark encounter was how Croatia's leadership had matured under cumulative exposure to stress.

Interviewee B noted that "repetition builds a mental script," explaining how crisis exposure itself becomes a form of training.[183] This observation connects directly to the DMDU notion of iterative learning loops, where uncertainty is not removed but normalized through repeated simulation.[184]

The match exemplified how leadership evolves across similar contexts by internalizing uncertainty rather than resisting it. In organizational terms, this equates to teams that operationalize ambiguity as part of the process rather than treating it as disruption.

4.3.1 Synthesis of Cluster 3

Across these five cases, leadership under pressure revealed the interaction between emotional steadiness, adaptive structure, and learned resilience. The interviews validated that decision-makers who frame uncertainty as procedural rather than exceptional maintain composure longer and act with greater clarity. Theoretical grounding from resilience theory and DMDU scholarship reinforces that adaptability and trust distribute the cognitive load of pressure, allowing leadership to scale from individual composure to collective confidence.[185]

4.4 Cluster 4 - Controversy and Unpredictability

This cluster explores moments when uncertainty was not driven by tactics or pressure alone, but by structural ambiguity: officiating bias, VAR controversies, and ethical

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[181] See *Wyscout*, Full Match Video: Russia vs Croatia, FIFA World Cup Quarter-Finals, 2018.
[182] See *Wyscout*, Match Report: Russia vs Croatia, FIFA World Cup Quarter-Finals, 2018.
[183] Interview with Interviewee B, conducted on 11.09.2025. See Appendix B.2.2.
[184] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 15.
[185] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 15.

uncertainty. These matches reflect decision breakdowns that occur when procedural clarity fails. In both football and business, such moments test not only decision-making speed but moral resilience, since fairness and transparency are as vital as competence. The parallels here extend to governance, compliance, and institutional credibility. As Adnan said, "what defines a strong system is not how it wins, but how it behaves when it is wrong."[186] His statement captures the moral dimension of uncertainty that is often overlooked in conventional analytics.

Case 1: Brazil vs Belgium (FIFA World Cup Quarter-Final, 2018)

This match is remembered for its tactical brilliance and its undercurrent of officiating ambiguity. Brazil's appeals for penalties and obstruction went largely ignored,[187] while Belgium thrived through rapid transition play. Beyond tactics, the psychological weight of perceived injustice shaped Brazil's composure. Neymar's visible frustration became symbolic of how emotional overreaction to systemic uncertainty can distort collective focus.[188] This interpretation is supported by reporting from The Sun, which highlighted the controversial VAR decision involving Vincent Kompany and Gabriel Jesus that went unpunished.[189]

Daniel noted that similar patterns appear in business settings, where any loss of trust in the process pushes leaders to rely on a more deliberate, structured way of thinking[190]. In both contexts, frustration shifts focus from performance to perceived fairness. Ethical decision theory identifies this as a loss of procedural trust, where outcomes lose legitimacy regardless of accuracy.[191]

In DMDU terms, this situation represents the failure of institutional adaptability. When the governing system fails to absorb unpredictable outcomes fairly, agents resort to emotional reasoning instead of strategic logic.[192] Brazil's leadership could not recalibrate,

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[186] *Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.*
[187] See *Wyscout*, Full Match Video: Brazil vs Belgium, FIFA World Cup Quarter-Final, 2018.
[188] See *Wyscout*, Full Match Video: Brazil vs Belgium, FIFA World Cup Quarter-Final, 2018.
[189] See *The Sun*, "World Cup 2018: VAR controversy as Brazil are denied clear penalty," 6 July 2018, no page number.
[190] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.
[191] See *Chan, C.; Ananthram, S.*, "A Neo-Institutional Perspective on Ethical Decision-Making," 2020, p. 7.
[192] See *Marchau et al.*, Decision Making under Deep Uncertainty, p. 15.

revealing that trust in fairness is a cognitive stabilizer in both football and organizational systems.

Case 2: Hungary vs Portugal (UEFA Euro Group Stage, 2016)

A chaotic and emotionally charged game that demonstrated volatility as both opportunity and risk. Portugal equalized three times to secure qualification, while Hungary's fluid attack repeatedly exposed Portugal's defensive disorganization. [193] The match lacked any discernible pattern, turning into an open-ended exchange shaped by instinct rather than structure.[194]

Manzar described such volatility as the test of whether you are reacting to noise or reading signals.[195] The unpredictability of the match mirrored environments where excessive reactivity replaces strategic patience. From a systems perspective, this game illustrates what complexity scholars call stochastic probability: when feedback loops overwhelm rational control.[196]

For business parallels, this kind of environment is seen during market shocks or social media crises, where leaders must distinguish between transient chaos and genuine trend shifts. Portugal's recovery through individual brilliance, especially Ronaldo's adaptive positioning, reflected resilience through decentralized decision-making rather than rigid control.[197]

Case 3: Liverpool vs Manchester City (Carabao Cup Final, 2016)

This match was notable for its fairness dilemmas and the thin line between composure and chaos in penalty shootouts. Several refereeing inconsistencies created frustration for both teams.[198] Ultimately, City triumphed through tactical control and mental recovery after missed chances.[199] The encounter emphasized the moral load leaders carry when decisions appear externally unfair but must still be managed internally.

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[193] See *Wyscout*, Match Report: Hungary vs Portugal, UEFA Euro Group Stage, 2016.
[194] See *Wyscout*, Full Match Video: Hungary vs Portugal, UEFA Euro Group Stage, 2016.
[195] Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.
[196] See *Kay, J.; King, M.*, Radical Uncertainty, 2020, p. 22.
[197] See *Wyscout*, Full Match Video: Hungary vs Portugal, UEFA Euro Group Stage, 2016.
[198] See *Wyscout*, Full Match Video: Liverpool vs Manchester City, Carabao Cup Final, 2016.
[199] See *Wyscout*, Match Report: Liverpool vs Manchester City, Carabao Cup Final, 2016.

Interviewee B mentioned during the interviews that when rules feel inconsistent, the team looks at leaders for emotional validation.[200] His observation fits within organizational behaviour theory that sees fairness as a precursor to motivation rather than its byproduct.[201]

Football's fairness dilemma mirrors how corporate ethics functions under imperfect regulation. Managers may disagree with governance processes but must preserve performance integrity until structural reform arrives. The lesson is that control and compliance are separate tasks: one stabilizes emotion, the other preserves order.[202]

Case 4: Manchester City vs Chelsea (UEFA Champions League Final, 2021)

This final encapsulated strategic misjudgement and ethical restraint. Pep Guardiola's unexpected decision to field no defensive midfielder against Chelsea's counterattack-oriented structure remains a case study in overconfidence.[203] The absence of tactical balance, coupled with a few officiating grey zones, tilted the psychological flow. Chelsea capitalized on this imbalance through composure rather than chaos.[204]

Adnan connected this to decision arrogance in corporate contexts, Leaders sometimes overcorrect success by proving that they can win on innovation alone, ignoring stability.[205] The match aligns with literature on bounded rationality, where excessive confidence narrows perceptual bandwidth and excludes disconfirming evidence.[206]

The managerial takeaway is humility as a form of cognitive discipline. Ethical and tactical restraint becomes an act of self-awareness. In DMDU framing, this is the adaptive checkpoint recognizing when to revert to stability over experimentation.[207]

Case 5: Real Madrid vs Juventus (UEFA Champions League Semi-Final, 2015)

A tense match dominated by officiating debates and psychological strain.[208] Madrid's penalty, awarded to Cristiano Ronaldo,[209] became a lightning rod for fairness discussions.

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200 Interview with Interviewee B, conducted on 11.09.2025. See Appendix B.2.2.
201 See *Kwakkel, J.H.; Haasnoot, M.*, Supporting DMDU, 2019, pp. 366-370.
202 See *Gabrys et al.*, The Influence of Substitution Decisions, 2025, pp. 4-11.
203 See *Wyscout*, Match Report: Manchester City vs Chelsea, UEFA Champions League Final, 2021.
204 See *Wyscout*, Full Match Video: Manchester City vs Chelsea, UEFA Champions League Final, 2021.
205 Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.
206 See *Kay, J.; King, M.*, Radical Uncertainty, 2020, pp. 120-123.
207 See *Walker et al*., Dynamic Adaptive Planning, 2019, pp. 54-56
208 See *The Guardian*, "Marcelo shines but Juventus defeats Real Madrid," 13 May 2015, no page number.
209 See *Wyscout*, Match Report: Real Madrid vs Juventus, UEFA Champions League Semi-Final, 2015.

Juventus, led by Gianluigi Buffon and Andrea Pirlo, displayed tactical containment and ethical composure despite emotional provocation. The team's focus under perceived injustice showed resilience grounded in procedural trust rather than reactive emotion.[210]

Interviewee A observed that the moment you start fighting the system instead of playing your role, you stop performing.[211] This encapsulates the ethical dilemma faced by organizations when regulation feels misaligned but must still be respected. The game therefore represents moral resilience, the ability to sustain fairness-driven discipline even when context rewards aggression.

Ethics scholars argue that this self-restraint differentiates resilient systems from opportunistic one.[212] In both football and business, ethical decision-making sustains legitimacy longer than temporary advantage.

4.4.1 Synthesis of Cluster 4

Across these matches, controversy and unpredictability exposed the fragile relationship between fairness, trust, and adaptability. Ethical leadership emerges not from certainty but from how uncertainty is handled when processes fail. The interviews collectively reinforced that emotional composure and procedural integrity are interdependent. Adnan and Daniel emphasized that fairness acts as a stabilizing structure during volatility, while Manzar and Interviewee A underscored the importance of maintaining trust even when authority seems compromised.

In synthesis, this cluster demonstrates that ethics is not peripheral to decision science but embedded within its stability. Just as teams adapt tactically to randomness, organizations must adapt morally to institutional imperfection. Both rely on the same cognitive resource, discipline under ambiguity.

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[210] See *Managing Madrid*, "Real Madrid and Juventus' midfields: Analyzing the tactical battle," 7 May 2015, no page number.
[211] Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.
[212] See *Bloemen, P.J.T.M., et al.*, Adaptive Delta Management, 2019, p. 323.

5 Discussion

This chapter synthesizes the findings from the comparative case analyses and interprets them in relation to the research objectives. It examines cross-case patterns of decision behaviour under uncertainty, consolidates these insights into the proposed Decision Intelligence Transfer Framework (DITF), and reflects critically on the limits and implications of the football-to-business analogy. The chapter concludes by outlining the theoretical contributions of the study and positioning the findings within the broader literature on decision-making under deep uncertainty.

5.1 Cross-Case Patterns

The comparative analysis of the four clusters revealed distinct yet overlapping patterns of decision behaviour under uncertainty. Across crisis, adaptation, leadership, and controversy cases, both coaches and managers relied on similar cognitive mechanisms to interpret unfolding events, manage time pressure, and recover from disruptions. These parallels indicate that while football and business operate in different domains, the structure of uncertain decision environments produces convergent behavioural responses. To organize these insights, this section identifies four categories of uncertainty: tactical, situational, ethical, and cognitive.

5.1.1 Tactical Uncertainty

Tactical uncertainty involves environments in which multiple strategies are viable but the time available for evaluation is limited. In football matches such as Liverpool vs Dortmund (2016) or Croatia vs England (2018), tactical structures collapsed after key turning points.[213], [214] Decision-makers reacted through rapid restructuring rather than by calculating optimal outcomes. This pattern mirrors the principle of Robust Decision-Making (RDM) described by Lempert, which favours adaptability and satisficing over optimization when future conditions cannot be predicted.[215]

Interview participants confirmed that tactical choices often depend on recognition rather than computation. Daniel Jarzynski explained that "you cannot always analyse everything before acting; sometimes you recognize the pattern from previous projects and move fast

[213] See *Wyscout*, Match Report: Liverpool vs Dortmund, UEFA Europa League, 2016.
[214] See *Wyscout*, Match Report: Croatia vs England, FIFA World Cup Semi-Finals, 2018.
[215] See *Lempert et al.*, Robust Decision Making, 2019, pp. 26-27.

before it collapses."[216] His comment echoed the logic of the Recognition-Primed Decision (RPD) model.[217] Tactical uncertainty, therefore, is managed through experiential pattern-matching rather than through formal analysis.

The matches also demonstrated that flexible structures enable faster orientation and reconfiguration. Teams with defined contingency routines, such as Real Madrid under Zidane, responded to disruption with minimal confusion.[218] This supports the Dynamic Adaptive Policy Pathways (DAPP) framework, which emphasizes maintaining clear trigger points and adaptive pathways to respond as events unfold.[219] Tactical uncertainty in business similarly rewards systems that anticipate failure points and pre-design shifts rather than relying solely on planning.

5.1.2 Situational Uncertainty

Situational uncertainty arises when environmental volatility or incomplete information disrupts established plans. It is the most visible form of uncertainty in both football and business contexts. In matches such as Brazil vs Germany (2014) or Manchester City vs Real Madrid (2022), entire tactical systems had to be reinterpreted mid-game.[220], [221] Coaches reframed strategies not because data suggested it but because the context demanded it. While not part of the formal case study selection, these matches serve as illustrative examples of situational uncertainty in elite football.

Interview insights reinforced this behaviour. Adnan Mushtaq noted that a good decision-maker reads the situation more than the data,[222] aligning with the Info-Gap Decision Theory (IGDT) approach, which frames decisions around maximizing robustness to surprise rather than minimizing expected loss.[223] The interviewees repeatedly emphasized the importance of context sensitivity and emotional composure, particularly when the available information is contradictory or delayed.

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[216] *Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.*
[217] See *Klein. G.*, Recognition-Primed Decision Model, 1993, p. 145.
[218] See *Managing Madrid*, "Tactical Review: 2016 Champions League Final," 29 May 2016, no page number.
[219] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, p. 72.
[220] See *Spielverlagerung*, "Brazil - Germany 1:7," 9 July 2014, no page number.
[221] See *Managing Madrid*, "An analytical breakdown of Real Madrid's win over Manchester City," 6 May 2022, no page number.
[222] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.
[223] See Ben-Haim, Info-Gap Decision Theory, 2019, pp. 96-98.

In football, this kind of uncertainty is amplified by momentum shifts and referee influence. In business, it is triggered by changing market data, regulatory updates, or competitor actions. Both contexts show that high-performing decision systems prioritize continuous observation and interpretation rather than one-time forecasting. The interviews highlighted that small signals like tone shifts in senior discussions or unspoken discomfort in the room often indicate when a strategic change is required. This reflects the orientation stage of the OODA Loop,[224] where sense-making precedes action.

5.1.3 Ethical Uncertainty

Ethical uncertainty concerns the integrity and legitimacy of decisions under ambiguous rules. Matches characterized by controversial VAR calls or subjective interpretations of fairness highlighted how moral ambiguity interacts with strategic decision-making. The discussion around fairness in these contexts parallels organizational decision ethics, where transparency, justification, and proportionality become essential.

Matthew Sinnicks critiques the tendency to romanticize competition by equating business with sport, arguing that business actions affect involuntary participants, unlike sport, where risk is voluntarily assumed.[225] This observation resonates with how football controversies create broader reflections on institutional accountability. During interviews, Manzar Shaikh commented that "transparency in how you arrive at a decision often matters more than the decision itself,"[226] a statement that aligns with Sinnicks' perspective on virtue and intention. From the DMDU perspective, ethical uncertainty can be interpreted as a governance problem rather than a technical one.

5.1.4 Cognitive Uncertainty

Cognitive uncertainty reflects the human limits of attention, interpretation, and emotional control under stress. The interviews repeatedly emphasized this dimension. Adnan Mushtaq described how over-thinking can be worse than acting too early,[227] while Daniel Jarzynski highlighted fatigue from constant simulation of outcomes during high-pressure

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[224] See *Horney, N.*, OODA Loop, 2011, p. 3.
[225] See *Sinnicks, M.*, Business-Sport Analogy Critique, 2022, p. 50.
[226] *Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.*
[227] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

decision phases.[228] Both accounts align with Gary Klein's argument that experience-based intuition compensates for bounded rationality in fast-moving situations.[229]

In the context of football, cognitive uncertainty manifests in penalty shootouts or late-game tactical choices where emotions and perception distort reasoning. This connects to the Adaptive Planning literature, which emphasizes iterative learning and mental model revision as central to navigating evolving uncertainty[230]. Cognitive uncertainty is thus the bridge between rational and experiential domains, showing how adaptability depends on the human capacity to balance emotion with analytical awareness.

5.1.5 Summary

Across the four uncertainty categories, the findings converge on one key insight: effective decision-making does not depend on eliminating uncertainty but on recognizing its type and responding appropriately. Tactical uncertainty rewards structured flexibility; situational uncertainty rewards contextual awareness; ethical uncertainty requires transparency and justification; and cognitive uncertainty relies on reflection and emotional regulation. Together, these behaviours describe what this thesis terms Adaptive Intelligence, a synthesis of robust design, situational awareness, and human judgment.

5.2 Decision Intelligence Transfer Framework (DITF)

The Decision Intelligence Transfer Framework (DITF) is introduced as the integrative outcome of this research, addressing the need to bridge formal decision science with real-time cognitive insight in uncertain environments. This framework emerged from the cross-case analysis of high-stakes football matches and their mapped business analogues, which revealed common patterns in how effective decisions are made under pressure. Rather than replicating surface-level tactics, the DITF transfers the underlying decision mindsets from sport to business emphasizing anticipation of uncertainty, adaptive teamwork, and continuous learning over static plans. By aligning the structural rigor of deep-uncertainty methods with the intuitive dynamism of on-field decision-making, the framework provides a coherent model for navigating complexity that is grounded in both theoretical and practical insights.

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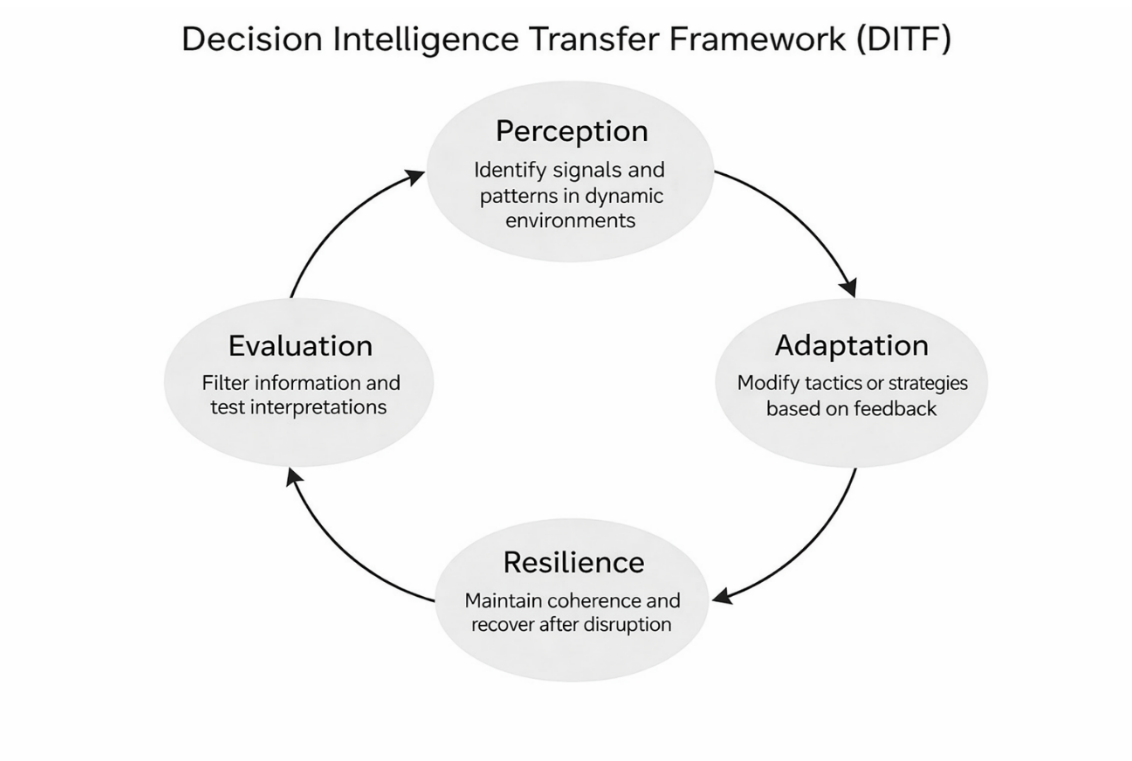
[228] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.
[229] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 143-144.
[230] See Walker et al., Dynamic Adaptive Planning, 2019, pp. 357-359.

At its core, the DITF demonstrates how acute situational awareness, tactical adaptability, and recovery mechanisms observed in football can directly enhance decision processes in business settings. It underscores that in turbulent contexts, decision quality depends more on contextual timing and coordinated judgment than on any precise prediction of outcomes. This perspective echoes the argument that navigating radical uncertainty requires narrative reasoning and adaptive interpretation rather than reliance on probabilistic models[231]. In essence, the framework reframes uncertainty not as a problem to eliminate, but as a condition to be continuously navigated through learning and flexibility.

Figure 2: Decision Intelligence Transfer Framework (DITF)



Source: Own Representation

Figure 2: The Decision Intelligence Transfer Framework (DITF) illustrates a cyclical process for navigating complex and volatile environments. It begins with Perception, where signals and patterns are identified to build situational awareness. This feeds into

[231] See *Kay, J.; King, M.*, Radical Uncertainty, 2020, pp. 157-159.

Adaptation, which involves modifying tactics or strategies in response to feedback and changing conditions. Resilience follows, emphasizing the ability to maintain coherence and recover from disruption. Finally, Evaluation ensures that information is filtered and interpretations are tested, enabling continuous learning. Together, these components form an iterative loop that supports agile decision-making under uncertainty. The framework is designed to be transferable across domains such as sport, business, and crisis management.

The DITF is structured as a cyclical process with four interlinked stages: Perception, Adaptation, Resilience, and Evaluation. Each component reflects a phase of decision-making identified in the research and is defined as follows:

5.2.1 Perception

Perception is the initial stage, focusing on scanning and interpreting the environment to build situational awareness. In a football context this means reading the game,for example, a coach and players identifying opponents' formations, shifts in momentum, or emerging patterns of play. Analogously in business, perception involves monitoring relevant signals such as market trends, competitor moves, or operational anomalies to discern the current situation. By identifying meaningful patterns amid chaos, decision-makers establish a knowledge base that informs subsequent choices and actions in the face of uncertainty.

5.2.2 Adaptation

Adaptation builds on what is perceived, involving the modification of tactics or strategies in response to feedback and changing conditions. On the pitch, this could entail a manager altering the team's formation, making a substitution, or shifting style of play as the match evolves and new information emerges (e.g. an opponent's weakness or a change due to a scored goal). In a business scenario, adaptation might mean pivoting a project plan, reallocating resources, or adjusting a strategy in real time when market conditions shift or an unexpected obstacle arises. This component captures the capacity to respond flexibly and decisively, ensuring that decision-makers can update their approach as the context evolves.

5.2.3 Resilience

Resilience is the third component, emphasizing the ability to maintain coherence and recover quickly after disruptions. A football team displays resilience when it regains focus and sticks to a clear game plan immediately after a setback such as conceding a goal or losing a player to a red card rather than unravelling under pressure. In business, resilience manifests in an organization's capacity to sustain its critical operations and morale through crises or shocks (for example, a sudden market downturn or a major client loss) and to bounce back with minimal long-term damage. This stage underlines the importance of stability and continuity amid turbulence: it is about absorbing shocks, preserving core integrity, and preventing short-term failures from derailing long-term objectives.

5.2.4 Evaluation

Evaluation is the final stage of the cycle, in which outcomes are assessed, and lessons are extracted before the cycle repeats. In football, evaluation may take the form of post-match analyses, coaches and players reviewing which tactics succeeded or failed, and why as well as in-game reflection (such as halftime adjustments based on the first half's events). In business, this corresponds to debriefing after a decision or project, analysing performance data and feedback to understand what worked and what didn't. During Evaluation, information is filtered and interpretations are tested, enabling teams to learn from experience and refine their mental models. Crucially, this evaluative learning is fed back into the next cycle of perception and planning, ensuring that each decision episode improves the organization's capacity to handle uncertainty in the future.

All four components operate in concert as a repeating loop that supports agile decision-making in chaotic conditions. The DITF is deliberately domain-agnostic: although derived from football scenarios, its principles are designed to be transferable across domains, guiding a coach in a championship match just as well as a CEO in a market crisis. In sum, the framework turns football decision-making into a lens for examining organizational strategy, revealing that effective decision intelligence in both arenas arises from reasoning under pressure rather than from deterministic planning. This connection between sport and business underscores the thesis's central finding - that high-pressure conditions in sport can inform and improve how leaders navigate uncertainty in the business world. The DITF thus provides a structured yet flexible bridge between football-based insights and business decision-making, illustrating a unified model of adaptive intelligence in action.

5.3 Critical Reflections

The Decision Intelligence Transfer Framework (DITF) provides a useful structure for understanding how adaptive reasoning can operate under uncertainty. However, the model also exposes several conceptual and philosophical tensions that must be acknowledged. These limitations are not flaws of the framework itself but reminders that all analogies between sport and management have interpretive boundaries. Critical reflection here focuses on three dimensions: epistemological robustness, analogy validity, and the philosophical tension between rationality and unpredictability.

5.3.1 Epistemological Boundaries

At the core of decision science lies the challenge of epistemic humility: how much can we truly know in a system defined by deep uncertainty? C. Brown et al. note that even the most sophisticated modelling frameworks rely on assumptions about how uncertainty behaves, which may fail when underlying systems change.[232] The DITF inherits this problem by emphasizing perception and evaluation as stages that presume decision-makers can recognize meaningful patterns in chaotic environments. In this sense, the DITF must be understood as a practical, not predictive, framework. It supports awareness, responsiveness, and flexibility, but it cannot guarantee correctness.

Interview insights confirmed this epistemic modesty. Interviewee A remarked that "you cannot know if you are right, only if you are still alive in the decision."[233] His phrasing, though informal, captures the applied essence of robust decision-making: survival of coherence rather than pursuit of truth. This stance is compatible with Ben-Haim's Info-Gap Decision Theory, which defines robustness as the ability to maintain acceptable outcomes even when the model is wrong.[234]

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[232] See *Brown, C.*, et al., Decision Scaling, 2019, p.264.
[233] *Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.*
[234] See *Ben-Haim*, Info-Gap Decision Theory, 2019, pp. 96-97.

5.3.2 Validity of the Analogy

A second limitation concerns the analogy itself between football and organizational decision-making. While sport provides a vivid simulation of high-pressure decision environments, its structure differs fundamentally from business or public policy.[235] Football operates in a closed system with finite time, known participants, and clear outcomes. Business and governance, by contrast, unfold within open systems characterized by delayed feedback, ethical ambiguity, and distributed accountability.

Matthew Sinnicks criticizes such analogies for their potential to obscure moral complexity. He argues that while sport rewards strategic success regardless of consequence, business decisions affect involuntary participants who do not consent to the risks.[236] This philosophical objection matters because it highlights that leadership under uncertainty must consider ethical legitimacy in addition to performance resilience.

The interviews partially support Sinnicks' critique. Adnan Mushtaq reflected that in corporate settings, the referee is invisible,[237] implying that fairness and rule enforcement are often subjective or politically constrained. In football, constraints are explicit and uniformly applied. In organizations, rules evolve through negotiation, interpretation, and cultural bias. Therefore, while the DITF effectively captures decision structures, it does not automatically address the moral or institutional dimensions of those decisions.

5.3.3 Rationality, Emotion, and Control

A third tension lies in the boundary between rational and emotional reasoning. Decision intelligence presumes that cognitive and behavioural processes can be observed, modelled, and transferred. Yet, as Kwakkel, Marchau and Walker emphasize, uncertainty also resides within the decision-maker's psychology, not only in external data.[238] Football illustrates this vividly: composure, momentum, and crowd influence often shape the outcome more than tactical planning.

[235] See Sinnicks, M., Business-Sport Analogy Critique, 2022, p. 55.
[236] See Sinnicks, M., Business-Sport Analogy Critique, 2022, p. 53.
[237] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.
[238] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 59-62.

Interviewee B and Manzar Shaikh both described how collective calmness during a crisis can be more decisive than any formal process.[239], [240] Their accounts align with the concept of bounded rationality, where individuals make satisfactory rather than optimal decisions under pressure. This mirrors the Recognition-Primed Decision model.[241] The DITF assumes that such cognitive boundedness is a feature, not a flaw, of decision-making because intuition, when grounded in experience, enables faster recognition and reaction.

However, this raises a final philosophical question: can intuition be systematized without losing its essence? If decision intelligence depends on embodied experience, then its transferability across contexts is necessarily limited. The analogy to sport, while illuminating, risks simplifying this complexity by turning tacit expertise into explicit flowcharts.

5.3.4 Reflection on Robustness

Despite these limitations, the DITF remains epistemically robust because it is descriptive, not prescriptive. It does not claim to predict outcomes but to structure thought in environments where certainty is unattainable. As Popper argues, DMDU frameworks are valuable precisely because they prioritize learning, adaptation, and dialogue over fixed optimization.[242]

The interviews reinforce this interpretive robustness. Decision-makers across different domains described uncertainty not as a temporary inconvenience but as a permanent operating condition. As Manzar Shaikh observed, the most mature teams stop trying to remove uncertainty; they learn to perform inside it.[243] This view captures the philosophical core of decision intelligence: that the measure of a good decision is not control, but adaptability.

5.4 Theoretical Implications

The findings from this study extend both theoretical and practical understanding across two distinct yet connected domains: decision science and sports analytics. The Decision Intelligence Transfer Framework (DITF) positions decision-making as an adaptive,

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[239] Interview with Manzar Shaikh, conducted on 13.10.2025. See Appendix B.2.5.
[240] Interview with Interviewee B, conducted on 11.09.2025. See Appendix B.2.2.
[241] See *Klein. G.*, Recognition-Primed Decision Model, 1993, pp. 141-142.
[242] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, pp. 382-383.
[243] See *Shaikh, M.*., interview conducted on 13.10.2025.

iterative process that integrates cognitive, emotional, and structural components. It moves beyond technical optimization and reframes uncertainty as a condition that can be navigated through contextual intelligence rather than eliminated through data precision.

5.4.1 Contribution to Decision Science

Within the field of decision science, the DITF contributes to what can be termed contextualized adaptability. Traditional models such as Robust Decision-Making (RDM) and Dynamic Adaptive Policy Pathways (DAPP) operate primarily at policy or infrastructural levels, where uncertainty is treated as an environmental variable to be mitigated. The DITF extends this by embedding adaptability within the decision-makers' recognition. It recognizes that decisions unfold not only within uncertain systems but also within uncertain minds.

This cognitive contextualization aligns with Popper's argument that decision frameworks should reflect human learning rather than technical perfection.[244] By emphasizing Perception and Resilience as explicit stages, the DITF internalizes the dynamic qualities of DMDU while incorporating the human capacity for judgment, emotion, and experience.

Interview evidence supports this theoretical direction. Adnan Mushtaq explained that data gives you a sense of probability, but experience gives you a sense of rhythm.[245] This statement mirrors Lempert's view that robust decisions require integration of both analytic and experiential forms of knowledge. Similarly, Interviewee A highlighted that adaptability is not simply reaction but a trained form of anticipation,[246] which echoes the learning-loop principle in the Conceptual Model of Planned Adaptation.[247]

Through this cognitive extension, the DITF helps fill a conceptual gap in decision science: the absence of an interpretive bridge between system-level frameworks and individual reasoning. Where RDM and DAPP offer institutional resilience, the DITF adds a behavioural layer that clarifies how decision-makers actually perceive and adapt in real

[244] See *Popper, S.*, Reflections: DMDU and Public Policy for Uncertain Times, 2019, p. 383.
[245] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.
[246] Interview with Interviewee A, conducted on 17.09.2025. See Appendix B.2.3.
[247] See *Sowell, J.*, "A Conceptual Model of Planned Adaptation, 2019, p. 297.

time. It is understood here that it supports a more grounded epistemology of uncertainty, one that displays the relation between rational process and human imperfection.

5.4.2 Contribution to Sports Analytics

The second theoretical implication emerges within sports analytics, where quantitative models have historically dominated. Most analytical approaches in sport aim to predict outcomes or optimize tactics through statistical regression and performance modelling. However, this study reveals that the most critical decisions often occur in conditions where quantification fails. In such moments, decision-makers rely on perception, emotional regulation, and distributed communication.

The DITF situates sports analytics within the broader cognitive landscape of decision science. It reframes football not merely as a system of measurable variables but as an experimental ground for observing how humans act under uncertainty. As Hemez and van Burren's work on Info-Gap suggests, resilience emerges not from certainty but from the capacity to function effectively amid incomplete information.[248] This insight holds direct relevance for tactical and managerial decision-making in elite sport, where ambiguity is constant and time constraints are extreme.

The case studies and interview data converge on this point. Daniel described in-game leadership as navigating with blurred vision but still trusting the compass.[249] His phrasing encapsulates the synthesis between intuition and structure that the DITF models. Similarly, Daniel Jarzynski and Manzar Shaikh both connected leadership under pressure with the idea of distributed cognition that teams adapt faster when decision authority is shared across experienced actors. This reflects a shift from command-based models to participatory intelligence, consistent with contemporary views in organizational theory and adaptive systems.[250]

Thus, the DITF enriches sports analytics by reframing decision performance as a blend of data interpretation and cognitive adaptability. Instead of seeking perfect prediction, it encourages analysts and managers to focus on decision learning: how situational awareness, feedback loops, and resilience interact to shape consistent success under

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[248] See *Hemez, F. M., Van Buren, K. L.*, "Info-Gap (IG): Robust Design of a Mechanical Latch," 2019, p. 205.
[249] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.
[250] See *Kwakkel, J.H.; Haasnoot, M.*, Supporting DMDU, 2019, pp. 357-359.

variability. This echoes the principles of exploratory modelling, where multiple plausible futures are simulated not to forecast accuracy but to improve readiness

5.4.3 Integrative Insight

Taken together, these contributions signal a broader theoretical movement. The DITF shows that decision-making under uncertainty must integrate three intertwined logics: data logic (for analysis), cognitive logic (for sense-making), and emotional logic (for endurance). Where DMDU frameworks conceptualize robustness as resistance to failure, the DITF recasts it as coherence through adaptation. In this respect, the framework bridges two worlds: the analytical precision of decision science and the embodied cognition of human performance.

Ultimately, the model positions uncertainty not as a limitation but as an operating environment. In both science and sport, what distinguishes effective decision-makers is not the absence of error but the persistence of intelligent response. As Marchau et al. summarize, the test of robustness lies not in whether the plan survives, but in whether the system continues to learn.[251]

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[251] See *Marchau et al.*, Decision Making under Deep Uncertainty, pp. 6-8.

6 Conclusion and Recommendations

This chapter concludes the thesis by summarizing the key findings and reflecting on their implications. It discusses how the results contribute to business practice and sports strategy, outlines the limitations of the study, and identifies directions for future research. The chapter closes by consolidating the overall contribution of the thesis to understanding decision-making under uncertainty through structured analogical analysis.

6.1 Summary of Key Findings

This thesis set out to explore how decision-making under uncertainty can be better understood and applied through analogies drawn from football. The analysis combined twenty high stakes matches with five interviews, supported by literature from the Decision Making under Deep Uncertainty (DMDU) framework and cognitive models of decision behaviour. The resulting Decision Intelligence Transfer Framework (DITF) connects perception, evaluation, adaptation, and resilience as sequential yet iterative phases in uncertain environments.

The first key finding is that uncertainty does not diminish decision quality if it is recognized and structured. Across the match analyses, managers and players demonstrated the ability to act with limited information by maintaining flexible goals and adjusting strategies as new signals appeared. This corresponds with Lempert's concept of robustness, where success depends less on prediction and more on performance across multiple plausible futures.[252] Football provided a microcosm of this logic: teams that remained open to tactical redirection after setbacks recovered faster than those anchored to rigid plans.

The second key finding concerns adaptability as both a cognitive and structural phenomenon. In the interviews, Daniel Jarzynski emphasized that speed of coordination is more important than speed of action.[253] His observation aligns with Walker's theory of Dynamic Adaptive Planning, which identifies monitoring and feedback as the foundations of resilience.[254] The analysis showed that adaptability is not simply reactive

[252] See *Lempert et al.*, Robust Decision Making, 2019, pp. 24-26.
[253] Interview with Daniel Jarzynski, conducted on 08.09.2025. See Appendix B.2.1.
[254] See *Walker et al.*, Dynamic Adaptive Planning, 2019, pp. 55

flexibility but a practiced mindset, reinforced through shared understanding and constant communication.

A third finding involves the ethical dimension of decision-making. The “Controversy and Unpredictability” cluster illustrated how fairness and legitimacy influence perceived stability in uncertain systems. This reflects Sinnicks’ warning that analogies from sport must acknowledge moral asymmetries between games and organizational life.[255] The DITF integrates this by embedding Resilience as not only a technical outcome but also an ethical state a system’s capacity to maintain integrity after disruption.

Finally, the study finds that decision intelligence is not the pursuit of perfect control but the management of controlled adaptability. The DITF contributes a behavioural layer to DMDU, demonstrating that effective decision systems combine robust analytical planning with human intuition and moral judgment. Football, in this context, serves as both a metaphor and an empirical test for understanding how organizations can design, communicate, and recover within unpredictable environments.

6.2 Implications for Business Practice

The practical relevance of these findings extends directly to corporate, financial, and crisis-management settings. The first implication is the value of structured flexibility. Businesses often fail not because they make wrong decisions but because they make inflexible ones. The DITF encourages organizations to establish decision protocols that include predefined signposts and adaptive pathways, a principle derived from Dynamic Adaptive Policy Pathways.[256]

In practice, this means developing strategic options that can pivot under new information without undermining core objectives. Adnan Mushtaq’s interview insight that data cannot replace narrative underscores the need for human contextualization of analytics.[257] Corporate decision-makers can adopt this by combining quantitative monitoring with qualitative interpretation reviewing not just the numbers but also the assumptions driving them.

[255] See *Sinnicks, M.*, Business-Sport Analogy Critique, 2022, pp. 50-51.
[256] See *Haasnoot et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 72-89.
[257] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.

A second implication concerns team-level communication. The case studies revealed that recovery from crises depends on shared situational awareness rather than hierarchical control. Football managers who empowered distributed decision-making during chaotic matches achieved better outcomes, paralleling findings from resilience engineering and organizational learning. For corporate teams, this means cultivating an environment where information flow is multidirectional, and decision authority can shift dynamically based on proximity to critical information.

A third implication is the integration of ethical resilience into strategic processes. Organizations, like football teams, must anticipate that disruption will test not only their plans but also their values. By explicitly mapping ethical thresholds equivalent to "adaptation tipping points" in DAPP businesses can predefine how far they are willing to adjust before compromising core principles.[258] This approach strengthens credibility, transparency, and stakeholder trust in volatile contexts.

Finally, the DITF offers a practical template for decision rehearsal. Just as football teams simulate scenarios through tactical drills, organizations can run decision simulations to explore potential responses under uncertain conditions. These exercises promote the development of cognitive resilience, the ability to maintain coherence and creativity even when traditional planning tools fail.

In summary, the DITF bridges analytical rigor and human adaptability. It provides organizations with a framework for learning from disruption rather than fearing it. By aligning structured decision science with behavioural insight, the model transforms uncertainty from a threat into a continuous source of strategic evolution.

6.3 Implications for Sports Strategy

The Decision Intelligence Transfer Framework (DITF) holds direct implications for modern football coaching, tactical modelling, and performance analytics. The framework reframes decision-making not as isolated acts of instinct but as structured cognitive processes that integrate perception, evaluation, adaptation, and resilience.

From a coaching perspective, this understanding promotes a shift from reactive management to anticipatory decision design. Coaches who recognize uncertainty as an

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[258] See *Lawrence, J., et al.*, Dynamic Adaptive Policy Pathways, 2019, pp. 190-194.

inherent condition of play can embed adaptive signposts into their tactical strategies. Just as policy designers establish thresholds that trigger pre-planned adjustments, football coaches can define tactical inflection points such as formation shifts after a red card or the substitution of a key midfielder when intensity drops.

The framework also enhances the application of data strategy in football. Traditional performance analytics focus heavily on predictive models, such as expected goals (xG) and possession-based probability, but these remain insufficient under volatile, time-compressed conditions. By applying DMDU principles, analysts can reorient data interpretation from prediction to preparedness. As Lempert notes, robust decision-making focuses on what remains effective across multiple plausible futures rather than optimizing for a single one.[259] In football, this translates to preparing strategies that hold under multiple game states rather than pursuing a fixed optimal model.

Adnan Mushtaq emphasized that data should inform but never dictate the tempo.[260] His comment reflects the balance between statistical insight and cognitive freedom, a relationship central to the DITF. It supports the use of explainable AI tools that assist, rather than override, human intuition.[261] Tactical innovation can thus evolve through a partnership between machine learning models and human decision-makers, maintaining interpretability and accountability.

Finally, the DITF's focus on resilience introduces a new way of understanding mental composure and momentum. Coaches can train cognitive recovery as a tactical skill, much like positional transitions. In short, the DITF provides a cognitive-scientific foundation for tactical agility. It merges the analytical process of DMDU with the embodied intelligence of sport, showing that resilience and adaptability can be trained, not merely observed. Football strategy, when informed by this framework, becomes less about controlling the game and more about learning to perform within its uncertainty.

6.4 Limitations and Future Research Directions

While the DITF contributes a structured synthesis of decision intelligence across sport and business, it is necessary to acknowledge its methodological and conceptual

[259] See *Lempert et al.*, Robust Decision Making, 2019, p. 25.
[260] Interview with Adnan Mushtaq Karol, conducted on 10.10.2025. See Appendix B.2.4.
[261] See *Vaykole, N.D.*, Explainable AI in Football, 2024, p. 56.

limitations. The research relied primarily on qualitative analysis, structured interviews and case interpretations which, although rich in depth, limit generalizability. Future studies could expand this framework through mixed-methods research that quantifies behavioural dynamics and validates them statistically.

One immediate direction is the integration of AI-driven simulation. Reinforcement learning and decision-tree models can replicate adaptive pathways under simulated uncertainty. Applying these techniques to football and organizational contexts would allow researchers to visualize how decision resilience evolves in real time.

Another limitation lies in contextual bias. Most of the interviews were concentrated among managerial and analytical professionals in financial institutions, with analogies drawn to football. While this dual context strengthens the interdisciplinary bridge, it also constrains external validity. Broader datasets could include decisions from healthcare, logistics, or public governance domains where uncertainty manifests differently. Comparative analysis across sectors would reveal whether adaptive reasoning behaves universally or depends on institutional structure.

The DITF also assumes that decision-makers can identify and act upon cognitive triggers. Future studies could employ ethnographic or experimental methods to examine the psychological barriers that delay adaptive action.

Finally, football analytics itself presents a frontier for future research. Integrating qualitative coding from tactical narratives with quantitative event data would enable a more comprehensive modelling of decision ecosystems. By merging data from event detection models[262] with cognitive interview insights, scholars can develop hybrid frameworks that bridge narrative understanding with measurable tactical shifts.

In conclusion, the limitations identified here are opportunities for expansion rather than weaknesses. The DITF provides a strong conceptual foundation that future researchers can test, simulate, and scale. By combining computational modelling, behavioural psychology, and domain-specific expertise, the next stage of inquiry can transform decision intelligence from a qualitative interpretation into a measurable, predictive science, one that maintains human adaptability at its core.

[262] See Bischofberger, J., et al., Event Detection in Football, 2024, pp.4-11.

Closing Remarks

This thesis began with a simple analogy: that decision-making in football mirrors decision-making in business. What unfolded through the research was a deeper recognition that uncertainty is not an obstacle to be removed, but a condition to be understood, navigated, and even mastered. Across every match analysed and every interview conducted, the same truth emerged: clarity in chaos is not a product of prediction, but of preparation and perception. The Decision Intelligence Transfer Framework (DITF) stands as both a theoretical and practical response to that reality. It demonstrates that resilience grows from systems capable of learning while acting, from teams that balance data with intuition, and from leaders who adapt without losing direction.

In uniting decision science with the lived complexity of sport, this work challenges the boundaries of how intelligence is defined. It argues that decision quality is not measured by precision alone, but by coherence, judgment, and ethical stability under pressure. The synthesis of analytical structure and human adaptability outlined here marks a step toward a broader theory of decision intelligence, one that values the imperfections that make real-world decisions not only possible but deeply human.

Appendix

Appendix A: Coding Matrices

Appendix A.1: RQ1 -Signal and Structure in decision making

Table 1: Matrix for RQ1-Signals and Structure in decision making

Theme / Code	Meaning	Adnan	Daniel	Interviewee A	Interviewee B	Manzar	Observation / Parallel
KPI and Data-Based Signals	Reliance on internal metrics, reports, or compliance rules when making quick decisions.	☑	☑	○	☑	☑	Similar to teams analysing performance data before matches.
Time-Bound Structure	Decision urgency determined by project or regulatory deadlines.	○	☑	☑	○	☑	Reflects time-driven tactical adjustments during a

							match.
Source Credibility	Trust placed on senior or familiar information sources.	☑	☑	☑	○	○	Mirrors how football managers rely on experienced assistants for pitch feedback.
Intuition as a Complement	Using experience or instinct when data is incomplete.	☑	☑	☑	☑	☑	Comparable to in-game tactical improvisation.

Source: Own representation based on semi structured interview

Appendix A.2: RQ2 - High-Stakes Business Decisions

Table 2: Matrix for RQ2- High-Stakes Business Decisions

Theme / Code	Meaning	Adnan	Daniel	Interviewee A	Interviewee B	Manzar	Observation / Parallel
Transparency Under Pressure	Communicating openly about risks, issues, or errors when decisions must be taken fast.	☑	☑	☑	☑	○	Equivalent to players signalling breakdowns during match pressure.
Collaborative Validation	Consulting peers or seniors before finalising a decision.	☑	☑	☑	☑	☑	Similar to team huddles before set-piece execution.
Quick Trade-Offs	Weighing business risk vs compliance or customer impact under tight timelines.	○	☑	☑	○	☑	Reflects managers deciding between attacking or defending near full-time.

Source: Own representation based on semi structured interview

Appendix A.3 RQ3 - Turning Points and Impact

Table 3: Matrix for RQ3 - Turning Points and Impact

Theme / Code	Meaning	Adnan	Daniel	Interviewee A	Interviewee B	Manzar	Observation / Parallel
Resource Shifts / Personnel Changes	Adjusting internal resources after unexpected events (staff exit, policy change, etc.).	☑	☑	☑	☑	☑	Analogous to substituting players after injuries or red cards.
External Shocks	Reacting to sudden external triggers (market or regulatory).	☑	☑	☑	☑	☑	Matches changing tempo after opponent's goal.
Adaptation as Routine	Continuous recalibration of plans under volatility.	☑	☑	☑	☑	☑	Core to tactical adaptability in football.

Source: Own representation based on semi structured interview

Appendix A.4: RQ4 - Frameworks under Uncertainty

Table 4: Matrix for RQ4 - Frameworks under Uncertainty

Theme / Code	Meaning	Adnan	Daniel	Interviewee A	Interviewee B	Manzar	Observation / Parallel
Reliance on Intuition over Frameworks	Preference for experience when frameworks fail or are unclear.	☑	☑	☑	☑	☑	Similar to captains making on-field calls outside coach instructions.
Documentation & Learning Loops	Capturing decisions post-event to improve future frameworks.	☑	☑	☑	○	○	Reflects post-match tactical reviews.
Decision-Support Tools	Interest in systems or simulations to test possible outcomes.	☑	☑	☑	☑	☑	Mirrors analytical tools predicting match scenarios.

Source: Own representation based on semi structured interview

Appendix A.5: RQ5 - Decision-Support Systems

Table 5: Matrix for RQ5 - Decision-Support Systems

Theme / Code	Meaning	Adnan	Daniel	Interviewee A	Interviewee B	Manzar	Observation / Parallel
Strategic Analogies (Defence vs Attack)	Framing business moves as proactive or defensive tactics.	☑	☑	☑	☑	☑	Same as teams deciding between holding the line or pressing high.
Simulation as Future Tool	Belief that predictive simulation could aid planning and prioritisation.	☑	☑	☑	☑	☑	Direct alignment with tactical modelling in football.
Human-Centric Limits	Recognition that unpredictability can't be fully modelled.	☑	☑	☑	☑	☑	Acknowledges the "chaos" factor in both domains.

Source: Own representation based on semi structured interview

Appendix B: Interview Consent forms and transcripts

Appendix B.1: Consent Forms

Appendix B.1.1: Daniel Jarzynski - Consent Form

Interview Consent Form

Project Title: *Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business*
Researcher: Faiz Karol
Institution: FOM Hochschule für Oekonomie & Management

Purpose of the Interview:
I am conducting research for my Master's thesis, which explores decision-making under uncertainty by drawing parallels between football match scenarios and business contexts. Your insights will help provide practical validation and examples for the study.

Recording and Use of Data:

- With your consent, this interview will be **audio-recorded** for accuracy.
- The recording will be used solely for transcription and analysis in the thesis.
- Your responses may be **quoted in the thesis**, either anonymously or with your name/title (depending on your preference below).
- Data will not be shared outside this research project.

Confidentiality Options (please tick one):
☒ I consent to be quoted with my **name and role** in the thesis.
☐ I consent to be quoted **anonymously** (e.g., "Senior Manager, HSBC").

Voluntary Participation:
Your participation is voluntary. You may skip any question or withdraw at any time before the thesis submission without penalty.

Consent:
I confirm that I have read and understood the above information, and I voluntarily agree to participate in this interview.

Name of Participant: DANIEL JARZYNSKI
Signature: ________________ Date: 04.09.2025
Researcher's Signature: F. Karol Date: 08.09.2025

Appendix B.1.2: Interviewee B - Consent Form

Interview Consent Form

Project Title: *Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business*
Researcher: Faiz Karol
Institution: FOM Hochschule für Oekonomie & Management

Purpose of the Interview:
I am conducting research for my Master's thesis, which explores decision-making under uncertainty by drawing parallels between football match scenarios and business contexts. Your insights will help provide practical validation and examples for the study.

Recording and Use of Data:

- With your consent, this interview will be **audio-recorded** for accuracy.
- The recording will be used solely for transcription and analysis in the thesis.
- Your responses may be **quoted in the thesis**, either anonymously or with your name/title (depending on your preference below).
- Data will not be shared outside this research project.

Confidentiality Options (please tick one):
☐ I consent to be quoted with my **name and role** in the thesis.
☑ I consent to be quoted **anonymously** (e.g., "Senior Manager, HSBC").

Voluntary Participation:
Your participation is voluntary. You may skip any question or withdraw at any time before the thesis submission without penalty.

Consent:
I confirm that I have read and understood the above information, and I voluntarily agree to participate in this interview.

Name of Participant: [redacted]

Signature: [redacted] Date: 11.09.25

Researcher's Signature: F.Karol Date: 11.09.2025

Appendix B.1.3: Interviewee A - Consent Form

Interview Consent Form

Project Title: *Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business*
Researcher: Faiz Karol
Institution: FOM Hochschule für Oekonomie & Management

Purpose of the Interview:
I am conducting research for my Master's thesis, which explores decision-making under uncertainty by drawing parallels between football match scenarios and business contexts. Your insights will help provide practical validation and examples for the study.

Recording and Use of Data:

- With your consent, this interview will be **audio-recorded** for accuracy.
- The recording will be used solely for transcription and analysis in the thesis.
- Your responses may be **quoted in the thesis**, either anonymously or with your name/title (depending on your preference below).
- Data will not be shared outside this research project.

Confidentiality Options (please tick one):
☐ I consent to be quoted with my **name and role** in the thesis.
☑ I consent to be quoted **anonymously** (e.g., "Senior Manager, HSBC").

Voluntary Participation:
Your participation is voluntary. You may skip any question or withdraw at any time before the thesis submission without penalty.

Consent:
I confirm that I have read and understood the above information, and I voluntarily agree to participate in this interview.

Name of Participant: [redacted]

Signature: [redacted] Date: 08.10.2025

Researcher's Signature: F. Karol Date: 08.10.2025

Appendix B.1.4: Adnan Mushtaq Karol - Consent Form

Interview Consent Form

Project Title: *Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business*
Researcher: Faiz Karol
Institution: FOM Hochschule für Oekonomie & Management

Purpose of the Interview:
I am conducting research for my Master's thesis, which explores decision-making under uncertainty by drawing parallels between football match scenarios and business contexts. Your insights will help provide practical validation and examples for the study.

Recording and Use of Data:

- With your consent, this interview will be **audio-recorded** for accuracy.
- The recording will be used solely for transcription and analysis in the thesis.
- Your responses may be **quoted in the thesis**, either anonymously or with your name/title (depending on your preference below).
- Data will not be shared outside this research project.

Confidentiality Options (please tick one):
☒ I consent to be quoted with my **name and role** in the thesis.
☐ I consent to be quoted **anonymously** (e.g., "Senior Manager, HSBC").

Voluntary Participation:
Your participation is voluntary. You may skip any question or withdraw at any time before the thesis submission without penalty.

Consent:
I confirm that I have read and understood the above information, and I voluntarily agree to participate in this interview.

Name of Participant: Adnan Karol

Signature: AKarol Date: 05.10.2025

Researcher's Signature: F. Karol Date: 10.10.2025

Appendix B.1.5: Manzar Shaikh

Interview Consent Form

Project Title: *Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business*
Researcher: Faiz Karol
Institution: FOM Hochschule für Oekonomie & Management

Purpose of the Interview:
I am conducting research for my Master's thesis, which explores decision-making under uncertainty by drawing parallels between football match scenarios and business contexts. Your insights will help provide practical validation and examples for the study.

Recording and Use of Data:

- With your consent, this interview will be **audio-recorded** for accuracy.
- The recording will be used solely for transcription and analysis in the thesis.
- Your responses may be **quoted in the thesis,** either anonymously or with your name/title (depending on your preference below).
- Data will not be shared outside this research project.

Confidentiality Options (please tick one):
☒ I consent to be quoted with my **name and role** in the thesis.
☐ I consent to be quoted **anonymously** (e.g., "Senior Manager, HSBC").

Voluntary Participation:
Your participation is voluntary. You may skip any question or withdraw at any time before the thesis submission without penalty.

Consent:
I confirm that I have read and understood the above information, and I voluntarily agree to participate in this interview.

Name of Participant: Manzar Shaikh

Signature: Manzar Shaikh Date: 13.10.2025

Researcher's Signature: F. Karol Date: 13.10.2025

Appendix B.2: Interview Transcripts

Appendix B.2.1: Daniel Jarzynski - Transcript

Participant: Daniel Jarzynski

Role: Senior Manager, ALCM, HSBC Poland

Format: Online audio call

Consent: Granted verbally and recorded

Date Conducted: 8 September 2025

Interviewer: Thanks for joining, Daniel. Before we begin, I'll give you a quick overview. The thesis is about decision-making under deep uncertainty, and I'm comparing how people make fast decisions in business with how managers react under pressure in football. Because of your experience with MREL, LDR and tight regulatory cycles, I thought your input would help me understand how people navigate high-pressure situations.

Daniel: Sure, happy to help. It makes sense. In our work, timing is everything. If a decision is delayed at the wrong moment, it can turn into a reporting issue. Things move fast, and you need to respond fast too.

Interviewer: Before we start, how many years of experience do you have?

Daniel: Around eight years. And honestly, experience teaches you more than theory. If I could redo my student years, I'd choose more real-world exposure earlier. Experience is what shapes how you behave when you don't have full information.

Interviewer: When you have to make fast decisions under uncertainty, what do you rely on the most?

Daniel: People. When time is limited and you don't have the luxury to run a full assessment, the most reliable thing is to talk to someone who understands the topic and get an angle from them. KPIs help, but I usually want one solid opinion from someone who knows the area.

Interviewer: And when there's a lot of data coming in at the same time, how do you decide what to look at first?

Daniel: If it's messy, I structure it. I categorise it and then start with whatever gives me the most complete picture. Usually, it's the dataset with the most features because it lets me look at the situation from multiple sides. And I prioritise the data with the biggest materiality or the highest impact. When things get volatile, rapid restructuring during financial volatility mirrors tactical substitutions when momentum shifts unexpectedly, you focus on whatever matters the most in that moment and move from there.

Interviewer: Do you have an example where you had to make a high-stakes decision with incomplete information?

Daniel: Yes. Recently we had to assess materiality for different reports and decide very quickly where the risk was. So, we started with the biggest numbers, the highest exposure, the things that could cause the biggest issue if wrong. And the items we could verify clearly. You don't waste time on noise when the stakes are high.

Interviewer: What makes a decision feel especially critical in your role?

Daniel: The consequences. Financial, operational, reputational. When a decision could shift the operating model or affect the organisation, that's when it becomes critical. In those moments, I always look for another opinion. I like being challenged and seeing the same issue from another perspective. Uncertainty doesn't allow ego.

Interviewer: When you face uncertainty, do you rely on a framework or intuition?

Daniel: Intuition and experience. Frameworks are helpful, but when the time is limited, intuition takes over. And intuition isn't guessing, it's built on patterns. In general, decisions under stress are less about options and more about perception. What you see in that moment is more important than the theoretical list of options.

Interviewer: And when you don't know the background of a new project but still need to make decisions quickly?

Daniel: I look for examples of similar projects. If something has been done before, I want to understand the dependencies, the structure, what went well and what didn't. I learn from patterns. And I talk to people to fill the gaps.

Interviewer: If we had a predefined decision model or simulation tool, where do you think it would be most helpful?

Daniel: Honestly, everywhere. A lot of things here are still basic. Systems aren't fully connected, automation is limited, and there's not much advanced tooling. A tool could help, but only as support. Certainty does not exist in a running process. You act, you observe, you reweight. So, the tool should support decisions, not replace them.

Interviewer: One thing we spoke about in the thesis is whether football analogies can help with business decision-making. Does that resonate with you?

Daniel: Conceptually yes, practically with caution. Football gives the impression that two situations are identical, same statistics, same flow but the outcome can be completely different. That's because football has more unpredictability: players, emotions, pressure. In business, people behave more predictably because of processes. But as a thinking model, it works. And especially under pressure. Pressure amplifies clarity for people who have rehearsed their response rather than their plan. That's the same in football and business. If your mind is prepared, you're better under fire.

Interviewer: That actually matches my findings.

Daniel: And for leadership, it's the same story. You rarely have perfect visibility. Sometimes navigating with blurred vision but still trusting the compass is all you can do. You don't always see everything, but you know the direction. That's how a lot of decisions work in regulatory reporting.

Interviewer: Thanks so much, Daniel. This really helps. I'll send you the cleaned transcript once everything is final.

Daniel: Perfect. Happy to look at it.

Appendix B.2.2: Interviewee B - Transcript

Participant: Anonymous- Interviewee B

Role: Werkstudent, ALCM, HSBC Germany

Format: Online audio interview

Consent: Granted verbally and recorded (Opted to be anonymous)

Date Conducted: 11 September 2025

Interviewer: Thanks for taking the time. Before we start, I just want to confirm that the audio will only be used for transcription, and everything stays anonymous as you wanted.

Interviewee B: Yes, that's completely fine.

Interviewer: Great. So, your total experience in the corporate world is around two and a half years, right?

Interviewee B: Yes, about that. Mostly here, and this is where I actually started handling real responsibilities.

Interviewer: Perfect. So, my thesis looks at decision-making under uncertainty both in football and business. I'll give you situations and you just tell me naturally how you think through a decision.

Interviewee B: Sure, go ahead.

Interviewer: When you have to make a quick decision in an uncertain situation and you don't have the luxury to escalate, what do you rely on the most?

Interviewee B: Usually, I check how similar problems were handled before. Sometimes there are older documents or past reporting examples that give some direction. But I don't blindly follow them. I cross-check with my own understanding from university and finance. I need to be able to defend whatever I submit. If I have the time, I double-check everything to make sure the numbers are actually logical.

Interviewer: And when information comes from everywhere KPIs, old files, regulator updates, team feedback, how do you prioritise?

Interviewee B: I start with the practical documents first, like the SOPs or step-by-step guides. Those tell you how the report is actually prepared. After that, I look at the general guidelines if something feels outdated. And if something still feels unclear, I rely on my own economic reasoning rather than just copying. I've noticed that when rules feel unclear, people naturally start prioritising fairness, it's like they want the decision to feel fair before anything else. I also think that's true for me. If the rules feel vague, I try to make the decision feel reasonable and fair to everyone involved.

Interviewer: Have you faced a high-stake decision where you had to act quickly?

Interviewee B: Yes. Once during month-end reporting our main tool broke. I was basically the only person available who knew how to fix it. If we couldn't get it running, the rework afterwards would have been massive. I immediately informed everyone that there might be a delay, and then I reached out to IT. We worked together, I explained the economic logic behind the inputs, and he understood the technical side. That combination helped us isolate the issue.

Interviewer: When a situation like that happens, how do you prepare or recognise that the decision is critical?

Interviewee B: Usually, you can sense when the outcome has a big impact. And as soon as something looks off, I inform everyone early. At the same time, I try to get support from people who know the tool or the process better. It's more about bringing the right people together quickly than trying to solve it alone.

Interviewer: Have you seen situations where the direction of a project had to change suddenly?

Interviewee B: Yes, definitely. Sometimes you need a particular data input for a report, but for technical reasons you can't retrieve it. Then I look at the bigger picture, how much does the missing information actually affect the results? If it's manageable, sometimes you can approximate it using historical averages or default mappings. It's not perfect, but for an intermediate solution it can work, as long as seniors approve it.

Interviewer: When you face a completely new task, do you follow the framework or rely more on intuition?

Interviewee B: For new tasks I start with the manual. And then over time, once I understand how the report works, I trust my intuition more. In the beginning, I just follow the instructions. But when I know the logic behind it, how the numbers move, how the tool processes things, then I can improve the process or automate parts of it. But that only comes after understanding the basic steps.

Interviewer: Let’s say you have very little time and you need to submit something fast. Would you rather include all the data or only the most relevant data that you know is correct?

Interviewee B: I would go with the data I’m confident about. If I’m 80-90% sure based on my background knowledge, I include it. But I’d rather submit less and be certain than include something that doesn’t make sense. If I’m time-restricted, relevance and correctness matter more than completeness.

Interviewer: Now, imagine you had a tool that could simulate the outcomes of different decisions like giving you different possible paths. In what kind of situation would that be most valuable to you?

Interviewee B: In general, I think something like that would help with decisions where the consequences aren’t obvious. It would give a bit of certainty, because you’d understand the possible outcomes more clearly. In our daily work we already do something similar with stress-testing calculating different scenarios. But if I could simulate strategic decisions or policy choices, that would be more useful. Especially in situations where adaptability matters, because sometimes the situation keeps evolving and you need to adjust your actions along the way. A tool that helps you forecast the effect of those adjustments would be very helpful.

Interviewer: Last question. Do you think unpredictable business events and structured decision-making frameworks from football can be connected?

Interviewee B: I think there is overlap. In business you need frameworks for clarity and structure, especially in big companies. But at the same time, you also need flexibility, because not everything can be captured in a guideline. It’s always a balance. And honestly, repetition builds a mental script, so the more you’re exposed to situations whether in football or work, the faster you recognise patterns. That’s why some of these parallels make sense to me. They’re not perfect, but the thinking can be similar.

Interviewer: That was really helpful. Thank you so much.

Interviewee B: No problem. It was interesting to think about these things from both the business and football side. Let me know when your thesis is done.

Appendix B.2.3: Interviewee A - Transcript

Participant: Anonymous- Interviewee A

Role: Senior Manager, Recovery and Resolution planning, HSBC Germany.

Format: Online audio interview

Consent: Granted verbally and recorded (Opted to be anonymous)

Date Conducted: 17 September 2025

Interviewer: Hi, thanks for joining. I just wanted to ask if you'd be okay with me recording the audio, but I won't record the video. I'll only be transcribing the audio, if that's okay with you.

Interviewee A: That's fine.

Interviewer: Before we start may I ask how many years of corporate experience do you have?

Interviewee A: Around eleven to twelve years.

Interviewer: I don't even have half of that. So, I have around five research questions, and all my interview questions are wrapped around those. My first research question is basically about whether the signals and structures of decision-making under deep uncertainty in sports and business match or don't match. So, when you have to make a very quick decision under uncertainty, what kinds of signals or information do you rely on the most? KPIs? Intuition?

Interviewee A: It depends on the background and the nature of the situation, but for me the first thing is always the timeline. I look at by when a decision needs to be completed. If I know there's still time to rethink the decision, to adapt, it's easier. If the timeline doesn't allow that, then I need to decide immediately. And in those situations, I don't necessarily rely on KPIs. I look at the goal I want to achieve, that matters more to me than the KPI itself. In the thesis you wrote, I think you phrased it well, my judgement in uncertainty is driven by timelines rather than KPIs. That's true.

Interviewer: And when there's a lot of pressure and a lot of information flying in at once, how do you structure or prioritise what matters?

Interviewee A: It really depends on who is providing the information. If it's senior management, like Andreas or other trusted colleagues, I make sure I understand it. But if it's from people I don't really know or trust, I skip it. And yes, that means I take the risk of missing something important, but that's exactly how it works.

Interviewer: Do you have any example where you had to take a critical decision with limited information and very little time?

Interviewee A: Of course. Deadline days. You know your data has issues, you know it's not at the standard you want. You look for quick fixes to ease the situation. But in the end, we usually decide to be transparent, internally and with regulators. That's part of resolution reporting every year. We know the issues, we communicate them, and we work on them afterwards. When models start breaking apart, our best response comes from the people closest to the data. That's true, in those moments we rely heavily on the ones who understand the issue first-hand.

Interviewer: What makes a decision feel especially critical for you? Like when you're preparing for it, do you consult others, rely on intuition, or follow some framework?

Interviewee A: I always speak with the key people involved. But I don't ask an open question, I prepare a preliminary idea before I speak to them. I explain my thought process and ask if they agree or if there's another angle.

Interviewer: In your experience, what kind of events completely change the direction of a project or strategy?

Interviewee A: Stakeholder surprises. We once requested external resources for the LDR template. Everything was prepared, consultants booked, hotels booked. One day before, the committee chair realised consultants were coming and said we don't have the budget. That was it. We had to send them home. We had to resource internally and rebuild the whole approach without the experts we planned. These things completely change direction.

Interviewer: And what about regulatory shocks, like sudden ECB requests?

Interviewee A: Happens often. And then it's not about frameworks, it's about speed. You find the right person, you gather whatever exists, and you prepare. These decisions are never made alone; there are always several people involved. The biggest challenge is making everyone understand the issue fast enough. You cannot know if you are right, only if you are still alive in the decision. Survival and momentum matter more than perfect accuracy in such cases.

Interviewer: When you have to take a decision quickly in an uncertain situation, do you rely more on old frameworks that existed before you, or more on your intuition and experience?

Interviewee A: Depends on the framework. Some are good, some are bad. But honestly most aren't well developed. So yes, I rely more on intuition. After the fact, if intuition worked, I try to update or build a framework from that so next time it's more structured. Adaptability is not simply reaction but a trained form of anticipation. That's something I strongly believe.

Interviewer: And what do you do to make your decision systematic when time is extremely short?

Interviewee A: You can't always make it systematic. If someone asks you later how you made the decision, you might not be able to explain it. But if I notice patterns I follow more than once, I build them into a framework afterwards. That's how I make intuition repeatable.

Interviewer: If we had a decision-support tool or framework designed from your experience, where in your role would it help the most?

Interviewee A: In finance areas, quantitative risk especially. When you have hard data, you can set thresholds that automatically trigger decisions or at least suggest actions. That's the only place where automation actually works. Qualitative decisions are harder because interpretation varies too much.

Interviewer: If you personally had a tool that simulates outcomes of your decisions, where would you use it?

Interviewee A: Prioritisation. Conflicting deadlines are constant. A tool that simulates outcomes and tells me what should go first and what can wait would be extremely helpful.

Poker was similar I used to play some, and I followed statistics. A tool like that in poker would tell me when to raise or call. At work, it's the same, it would help with time and priority decisions.

Interviewer: And my last question. Do you think analogies between unpredictable business events and decision-making in football actually work, or are there limitations?

Interviewee A: I think generally they can be combined, but not in every area. The HR-type analogies don't always work. But the competitive pressure analogy you gave, like a strong competitor suddenly attacking your market share, that works. Pressure, reaction, whether you go defensive or attacking, that part aligns well between football and business. That's where I see the overlap.

Interviewer: Perfect. Thank you so much. That was actually insightful and a good interview. And I will share the transcript with you and will refer to you as anonymous as you opted.

Interviewee A: Sure. And put it anonymous, Interviewee A.

Send it to my HSBC email when you have it.

And yes, that was a good interview.

Appendix B.2.4: Adnan Mushtaq Karol - Transcript

Interview Transcript: Adnan Mushtaq

Participant: Adnan Mushtaq

Role: Data Scientist, SURU (LIXI Germany)

Format: Online audio interview

Consent: Granted verbally and recorded

Data Conducted: 10 October 2025

Interviewer: Before we begin, I want to confirm that you're okay with me recording this and using the transcript for my thesis.

Adnan: Yes, sure. You can record it.

Interviewer: Great. Could you briefly introduce yourself and your role?

Adnan: I work as a Data Scientist at SURU, which is part of LIXI in Germany. Our company focuses on water-damage prevention, so we deal with a huge amount of sensor and behavioural data. My job is to take that data, analyse it, build models and algorithms that detect potential damages, and create dashboards or insights for different stakeholders. I have around four and a half years of experience as a Data Scientist, and around six years if you include my working student period.

Interviewer: Perfect. The interview is for my thesis on tactical intelligence transfer from football to business. I'm trying to understand how people make quick, high-stakes decisions under uncertainty and whether the way we think in sport carries over into business. So, I want to start with signals. When you're making quick decisions and there's uncertainty, what do you rely on the most?

Adnan: The first thing is always the quality of the data. Before I even think about signals, KPIs, or options, I want to check whether the data is complete, accurate, consistent, and reliable. If the data quality is bad, then whatever model or approach I build will collapse. And when a model starts collapsing, the first instinct shouldn't be to discard it but to understand which variables are still working. That's something I've learned the hard way. After quality, it really depends on the context. But usually, I rely on a combination of internal KPIs, system logs, user behaviour patterns, and feedback. I try not to rely on gut feeling. I prefer to structure everything as a feedback loop: input - system - output - evaluation. That iteration helps me understand very quickly what' s outperforming and what' s underperforming.

Interviewer: And how do you prioritise incoming information when it's chaotic?

Adnan: Priority depends. I ask myself a mix of questions.

Who provided the data? Can I trust that source? Has it been reliable before? Is the data clearly defined? And then I check completeness, accuracy, and consistency.

You asked about intuition earlier, I think intuition is useful only when it's grounded in experience. A good decision-maker reads the situation more than the data, but that doesn't mean ignoring the data. It means understanding the context around it.

Interviewer: Do you have an example of a high-stakes decision where you didn't have all the information?

Adnan: One example is our app feedback system. We rely heavily on user feedback from the Play Store, App Store, and surveys. When feedback suddenly spikes or becomes negative, we need to make quick decisions about what to fix, what to roll back, or what to investigate. Sometimes the signals are contradictory, but we still have to act.

In those moments, over-thinking can be worse than acting too late, so we focus on the most credible signals and move.

Interviewer: What makes a decision feel critical to you?

Adnan: Anything that has a high impact money, safety, user trust, or long-term strategy. And especially decisions that are irreversible. When you can't roll something back, the pressure increases.

Also, decisions become critical when stakes are high and timelines are tight. When something goes wrong and we're unsure why, that's where the stakes really rise. And honestly, what defines a strong system is not how it wins, but how it behaves when it is wrong. That's when the real decision-making ability shows.

Interviewer: How do you prepare for those moments?

Adnan: First, I clarify the goal. Am I trying to minimise risk? Protect the company's reputation? Maintain momentum? This clarity matters.

Second, I identify what is known and what is unknown.

Third, I set a time box. If I have 30 minutes, I commit to deciding within those 30 minutes.

Then I use quick filters:

- What's the impact?

-What happens if I'm wrong?

-Is the decision reversible?

-Is speed more important than accuracy?

And finally, I try to get input from other data scientists or engineers. Knowledge should move around. Decisions become better when multiple brains pressure-test them.

Interviewer: What about team disruptions? Sudden staff changes, missing people, blocked tasks?

Adnan: That happens a lot. When someone critical becomes unavailable, I assess three things:

1. Task criticality - is it blocking someone else?
2. Skill uniqueness - can someone else realistically, do it?
3. Timing - can we wait, or is there a deadline?

If the skill is transferable, we shift it to another person. If we can wait, we wait. If it's blocking, we escalate. But in all of this, communication is key. We have monthly planning sessions for this reason, but emergencies still happen.

Interviewer: You mentioned earlier that football analogies made sense to you. Do you see similarities?

Adnan: Yes. In both places, uncertainty is normal. And you often have to adapt mid-game. For example, in model calibration, we sometimes modify data weights in flight, even if it temporarily breaks the model. That's similar to coaches shifting tactics during a match. And I believe true adaptability happens when both sides of the system are learning at the same speed, whether its players learning from opponents or teams learning from market behaviour. Also, structured adaptation is anticipatory, not reactive. You want to see the signs before the collapse happens. Even in football, the best managers change shape before the match flips.

Interviewer: That's exactly the type of thinking my thesis is exploring.

Adnan: Yes, and sometimes the principle is simple, it's like local optimization inside a global uncertainty. You focus on the small controllable things first, while the big picture keeps shifting. That's similar to how we tackle uncertain projects.

And the leadership aspect is also similar. I think leadership under pressure means pre-committing to calm. If the leader panics, the whole team panics. Even in data teams, people look at how the senior person reacts.

Interviewer: That's a great line.

Adnan: It's true. Even when visibility is terrible, sometimes leadership is navigating with blurred vision but still trusting the compass. You don't always have clarity; you just need direction.

Interviewer: If you had a tool that could simulate outcomes of different choices, where would it help the most?

Adnan: Scenario comparison. If a tool can simulate how each alternative behaves under different uncertainties, that would help a lot. It would help us compare strategies side by side, highlight risks, and speed up decision-making. And it would complement our judgement, not replace it. Because data should inform the tempo, but it should never dictate it.

Interviewer: That ties perfectly to the decision-intelligence angle. Before we wrap up, do the analogies we discussed resonate with your experience?

Adnan: Yes, absolutely. Everything we discussed about risk, timelines, feedback loops, adaptation, these are things we deal with every day. Your questions made me think about them in a structured way. And in the end, data gives you probability, but experience gives you rhythm. That's why sport analogies make sense to certain extent, experience shapes how we behave under pressure.

Interviewer: Thank you so much, Adnan. I'll send you the cleaned transcript and summary as required.

Adnan: Perfect. Glad to help.

Appendix B.2.5: Manzar Shaikh - Transcript

Participant: Manzar Shaikh

Role: Assistant Vice President and mobile app product lead at Fatakpay, India.

Format: Online audio interview

Consent: Granted verbally and recorded

Date Conducted: 13 October 2025

Interviewer: So, my thesis is about decision-making under uncertainty. I compare football turning points with unpredictable business events to see if the strategic thinking overlaps. Before we start, can you tell me your role?

Manzar: I'm the Assistant Vice President and mobile app developer in FatakPay.

Interviewer: And your total experience?

Manzar: Around five and a half years.

Interviewer: When you have to make very fast decisions in uncertain situations. compliance changes, sudden issues, unexpected problems, what signals do you rely on the most? KPIs, competitor signals, team feedback, or intuition?

Manzar: Mainly KPIs and internal discussions. Those two guide everything. Of course, in tech you also use instinct when you don't have time. And honestly, sometimes being small forces clarity. When you're a small team, you immediately feel the impact of a decision, so you focus faster.

Interviewer: And when a lot of data comes at once, how do you prioritise it?

Manzar: I map everything back to KPIs. If I have three different inputs, I link each one to the key metrics. Whatever affects the main KPI the most becomes priority. That's how I filter noise.

Interviewer: Do you have an example where you had to make a high-stakes decision very quickly with incomplete information?

Manzar: Yes. Our app got removed from the Play Store because of an RBI SMS compliance issue. We had lakhs of customers, huge campaigns running, and suddenly no one could download the app. We had minutes, not days. The choice was whether the SMS data was more important or getting the app back online. We decided the app had to go back first; we could rebuild the SMS logic later. That was one of those moments where you break everything into small tasks and solve them in order.

Interviewer: How do you usually prepare for critical decisions?

Manzar: You really can't prepare everything, this field changes too fast. You stay mentally ready and rely on experience. You also talk to others. This isn't individual work; it's a team effort. And leadership under pressure exposes where natural leaders assume control when formal leadership is compromised. That happens here too. You see who actually stabilises things when something blows up.

Interviewer: In your experience in fintech, what kinds of events completely change the direction of a project or strategy?

Manzar: Fintech is volatile. RBI can suddenly change a rule. Google Play can enforce something new. Market results can shift overnight. Staff changes can disrupt flow. You always keep track of what's a grey area, black area, white area. You try to stay as close to compliant as possible so when something hits, you can adapt without collapsing.

Interviewer: And when something sudden happens, like an RBI guideline change, how do you assess impact?

Manzar: We check every touchpoint that will be affected. What user drop we'll face, monetary loss, feature breakage, and what needs to be rebuilt. Then we adjust gradually, this industry changes every month.

Interviewer: Do you follow a standard framework for handling uncertainty, or rely on experience?

Manzar: We use Scrum as the structure. But actual decisions are based on experience, instinct, and alignment with the team.

Interviewer: What helps you make decisions more systematically when time is short?

Manzar: Validation from colleagues or business owners. It's not a solo game. If the team aligns, the decision is stronger.

Interviewer: If you had a decision-support system. a tool to show potential outcomes, where would it help you the most?

Manzar: Regulatory reporting, audits, compliance checks, and situations like when the Play Store removed our app. Anywhere you need to double-check risk.

Interviewer: And if it could simulate outcomes of different choices?

Manzar: I'd use it for strategic planning, team management, risk management, time management. Those areas benefit the most.

Interviewer: Finally, does the football analogy resonate with your experience?

Manzar: Yes. Business also has unpredictable events, and you still need a structured way of reacting. And honestly, transparency in how you arrive at a decision often matters more than the decision itself and the more you face crises, the faster you react next time. It can be useful in the future, even if people need time to adapt to the idea.

Interviewer: Perfect. Thank you so much, Manzar.

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Declaration in lieu of oath

I, Faiz Nizamuddin Karol, solemnly declared that the report titled "Decision-Making Under Uncertainty: Tactical Intelligence Transfer from Football to Business " is entirely my own work. I confirm that all the ideas, theories, opinions, and information presented within this report are original unless appropriately cited and referenced. I have diligently acknowledged all sources consulted during the preparation of this report to avoid any instances of plagiarism or academic dishonesty. Furthermore, I assert that I have not previously submitted this report, or had any similar version of it, for academic credit elsewhere. I fully understand the gravity of academic misconduct and accept full responsibility for the authenticity and integrity of this work. By affixing my signature below, I pledge my commitment to upholding the principles of academic honesty and excellence.

Essen, 02/01/2026 Faiz Nizamuddin Karol